

Data-Driven Diagnosis: A Machine Learning Study on Cognitive Impairment

**Dissertation Submitted for
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by

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JUNE – 2023**

CERTIFICATE

This is to certify that the dissertation entitled '**Data-Driven Diagnosis: A Machine Learning Study on Cognitive Impairment**' being submitted by Tharun Kota, Registration No: 21mslsbf16 to Central University of Punjab, Bathinda for the award of Master's degree in the M.sc Bioinformatics. (discipline/subject) is a bonafide research work carried out by her/him under my supervision. The results presented in this dissertation have not been submitted elsewhere for the award of any other degree. In my opinion, the dissertation work has reached the standard of fulfilling the requirements for the award of Master's degree in accordance with the regulations of central University of Punjab, Bathinda.

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DECLARATION

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Abbreviations:

Abbreviation	Full Form
AD	Alzheimer's Disease
MCI	Mild Cognitive Impairment
GEO	Gene Expression Omnibus
RFC	Random Forest Classifier
SVC	Support Vector Classifier
RFE	Recursive Feature Elimination
BAS	Balance Accuracy Score
GBD	Global Burden of Disease
ML	Machine learning

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Abstract:

Alzheimer's disease (AD) is a chronic neurodegenerative disorder representing the most prevalent form of dementia worldwide. Unfortunately, it remains an incurable condition that irreversibly damages brain cells. This study employed a machine learning approach to develop a robust algorithm that predicts the preclinical stage of Alzheimer's based on gene expression data. The research utilized gene expression data obtained from three distinct stages of AD, namely Normal, Mild Cognitive Impairment (MCI), and Alzheimer's. The dataset consisted of 329 samples, including 145 samples from the Alzheimer's stage, 80 from the MCI stage, and 104 from the Normal stage. To identify relevant genes, differentially expressed genes were selected using an adjusted p-value threshold of less than 0.05, with the exclusion of duplicated and genes with unknown functions among a total of 30,000 profiled genes. Subsequently, the study employed the Recursive Feature Elimination (RFE) technique in combination with Support Vector Classifier (SVC) to achieve satisfactory performance, with a balanced accuracy score (BAS) of 72%.

1. INTRODUCTION

Neurological disorders have a significant impact on global health. According to the Global Burden of Disease (GBD) Study, these disorders account for 3% of the worldwide disease burden. Despite the relatively small percentage, several neurological disorders such as dementia, epilepsy, migraine, and stroke rank among the top 50 causes of disability-adjusted life years (DALYs)(*Thakur et al.,2016*).

Alzheimer's disease (AD) is one of the most prevalent causes of dementia, characterized by developing memory loss and cognitive impairment worldwide(Schneider, 2017). Annually, there is a record of ten million fresh instances of dementia, and projections indicate that by 2050, roughly 135 million individuals will experience a certain level of dementia. (*Dementia*, n.d.)

Mild Cognitive Impairment (MCI) is recognized as the preclinical phase of Alzheimer's disease. Patients may exhibit some cognitive difficulties noticed by themselves and their family members. However, they can still carry out their daily routines, although they may require some support. It is important to note that only a tiny percentage of MCI patients, about 10-15%, progress to develop AD each year, while the majority remain stable cognitively. One of the significant stages in identifying Alzheimer's disease is MCI. It is crucial to detect MCI since taking necessary actions at this stage can prevent it from developing into AD.(*Khojaste-Sarakhsi et al., 2022*)

Detecting patients with MCI who are at risk of developing AD is crucial not just for prescribing appropriate disease-modifying drugs like aducanumab(S, 2021), but also for implementing a strategy of early detection and diagnosis. This approach allows enough time for non-pharmaceutical interventions to take effect and delay the onset or severity of symptoms. Such interventions include cognitive reserve maintenance and social stimulation therapy.(*Beheshti et al., 2022*)

In the past, diagnosing Alzheimer's disease (AD) relied mainly on brain MRI and neuropsychological testing. However, understanding the molecular-level

changes in AD has been challenging due to the difficulty of obtaining samples from the posterior brain of both normal individuals and AD patients. Fortunately, recent studies have generated extensive omics data from various brain regions, allowing the development of prediction methods using machine learning (ML) models to diagnose AD as early as possible.(El-Gawady et al., 2022)

The term Machine Learning (ML) refers to the field of computer science that uses algorithms and statistical models to enable machines to learn through patterns and inference without explicit programming. As machines know from experience, ML algorithms can improve their performance automatically. ML systems employ various techniques, train models, and determine the output autonomously.

The advantages of using such methods, including convenience and cost-effectiveness, and evidence suggests that they can outperform clinicians in some cases. Consequently, there has been significant research on applying ML to AD diagnosis using different medical data formats, such as MRI.

The machine learning model is pre-trained on the available data to detect a specific type of pattern using an algorithm in an ML system. That means it scrutinizes the data and uncovers the underlying patterns in the data. The dataset's feature scaling techniques and existing answers determine the formula that depends upon the input and output functions and uses it on novel data to predict the outcome. Hence, the algorithm uses data collection for training and formulating the data to predict the output and saves the procedure for future purposes.(Bari Antor et al., 2021)

The use of machine learning is being increasingly acknowledged to achieve better predictive capabilities by utilizing information from multiple predictors. This interdisciplinary field encompasses engineering, computer science and statistics, and enables computer-based learning through inference and patterns without explicit programming. By employing known training examples, machine learning algorithms can provide the most accurate predictions for new cases where the outcome is still unknown.

In order to find MCI and AD biomarkers and learn more about the biological mechanisms and molecular actions that contribute MCI to AD progression, gene expression data has been used. Transcriptomic data can be measured at the genome-wide level thanks to gene expression profiling. This kind of data analysis can provide light on the relationship between MCI and AD and its clinical symptoms and the gene connections that are crucial to the progression of AD.

Recent advances in genomics and gene expression data have substantially enhanced the capacity to collect genetic data from different organs and brain areas of the same individual. To ascertain the degree to which AD affects the cognitive system, gene expression data from several brain areas that are a component of the cognitive system must be integrated. This is crucial since AD advances sequentially and affects many different brain regions. Studies have discovered altered levels of gene expression in the AD-related white and grey matter.(Arzouni et al., 2020)

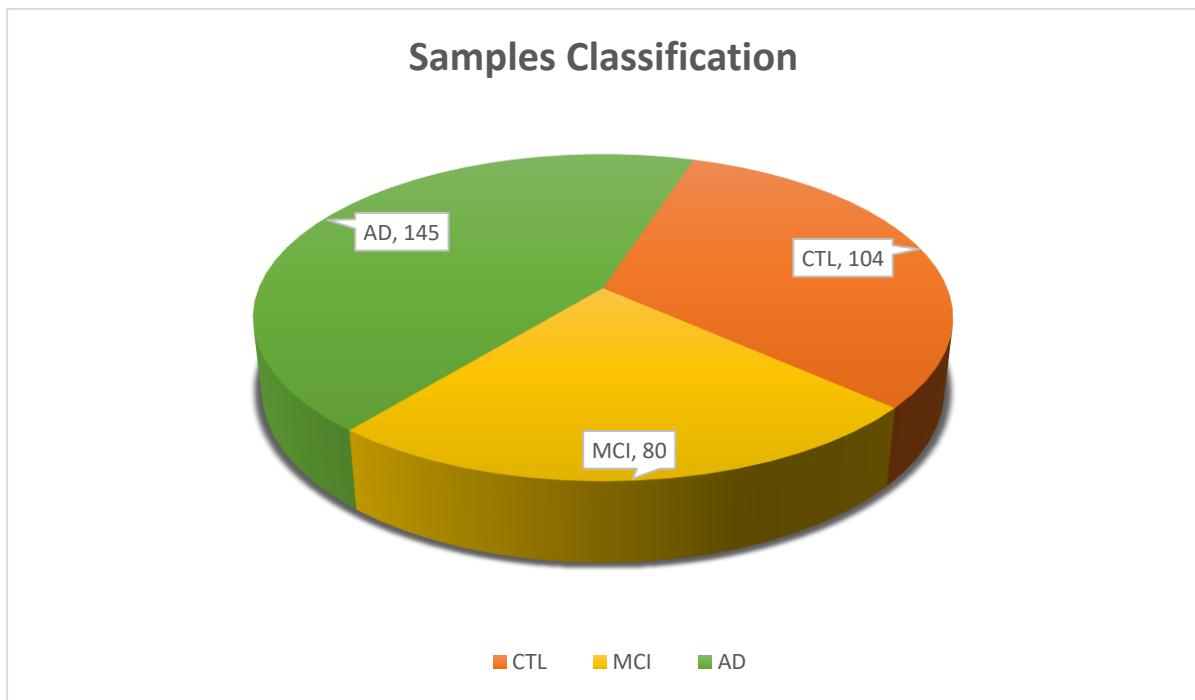
While gene expression studies have revealed important patterns, the majority of the data used in these studies came from biopsy or autopsy-based samples, which are not easily applicable to clinical settings. However, a small number of studies have used blood-based expression data to identify significant genes associated with AD or to predict early AD. These studies indicate that the levels of progranulin expression in the blood are elevated in AD and MCI.(Lee & Lee, 2020)

Over 50 published studies have applied machine learning techniques to predict MCI-AD conversion. Various combinations of predictors have been utilized along with different machine learning techniques to forecast the conversion from MCI to dementia, ranging from one year to five years after the baseline assessment. The results from these studies have varied, with some achieving performances just above chance and others showing high levels of accuracy.(Grassi et al., 2019)

2.MATERIALS & METHODS

2.1. Data Source:

To ensure the effectiveness of this study, it is essential to have a gene profiling database that includes a substantial number of samples from individuals with Mild Cognitive Impairment (MCI), Alzheimer's (AD), and healthy controls. We obtained the required data from the National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO) using specific criteria such as "Alzheimer's disease" and "brain" for the species *Homo sapiens*. From the available options, we selected the GSE63063(Sood et al., 2015) series, which consists of two platforms: GPL6947 and GPL10558. For our study, we focused on the GPL6947 platform, which encompasses a total of 329 samples. Among these, there are 145 samples from individuals in the Alzheimer's stage, 80 samples from individuals in the mild cognitive impairment (MCI) stage, and 104 samples from individuals in the normal stage (CTL).



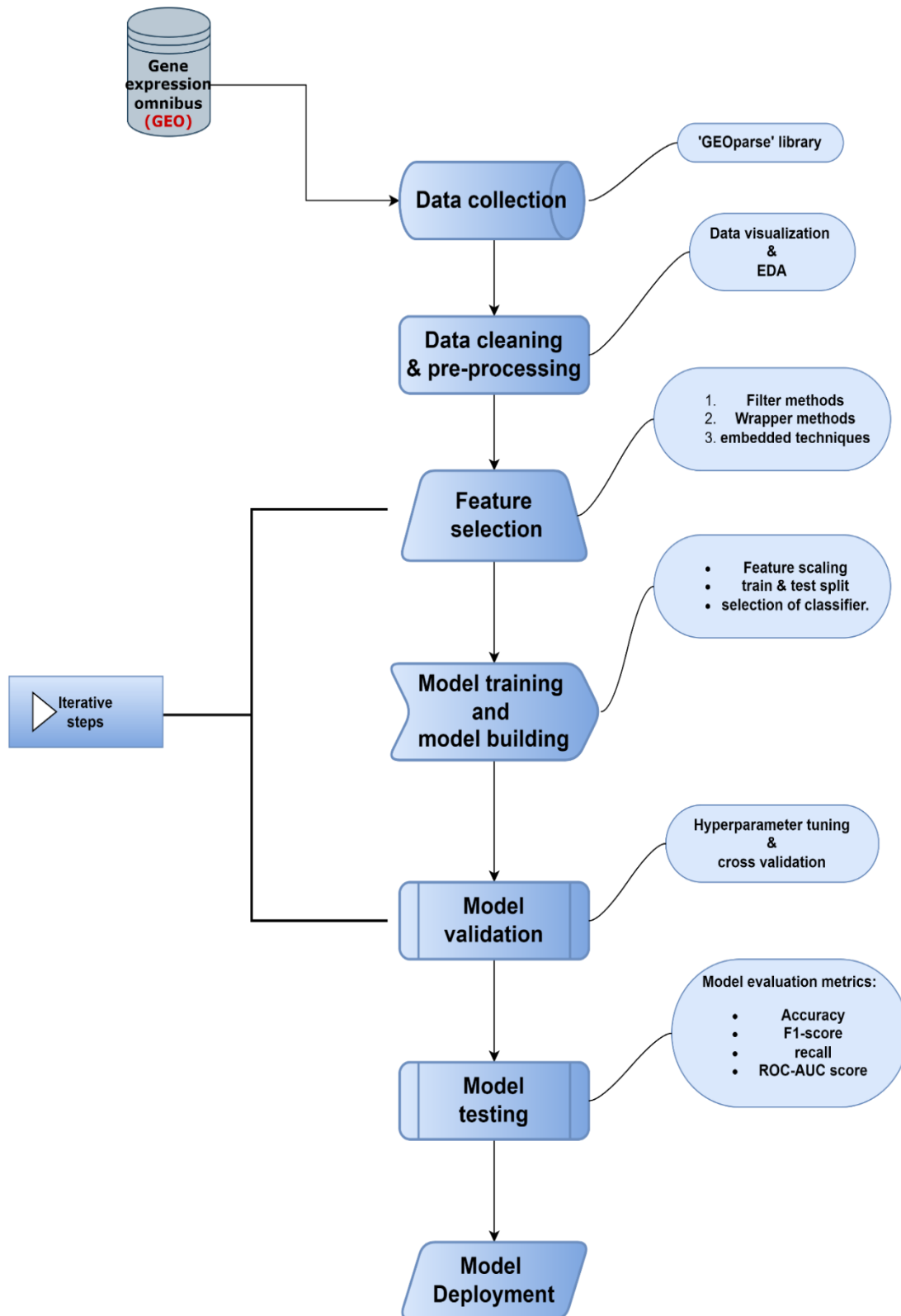
Fig_1: Pie graph representing samples in each class

2.2. Software used & Data extraction:

Software used for this study was Anaconda jupyter notebook version of 6.4.8 was registered to Tharun kota and was executed under the windows 11 operating system with AMD Ryzen 5,5500U with Radeon Graphics and 16.0 GB RAM. System type 64-bit operating system, And expression data of the GEO dataset GSE63063 was downloaded using 'GEOParse' library. GEOparse is python package that can be used to query and retrieve data from Gene Expression Omnibus database.

2.3. Iterative Cycle of Machine Learning:

Machine learning approach has few iterative steps as shown in the following fig. Iteration steps include feature selection and following by the rest as feature selection is most important i.e., genes in this case finding which genes are most important for creating a balanced decision boundary between three classes CTL, MCI&AD.



Fig_2: Machine learning methodology map for the current study

2.4. Data collection, cleaning & pre-processing: -

After retrieving data from the GEO database using the 'GEOparse' library, the acquired raw data is not immediately suitable for use and requires preliminary exploratory data analysis. This analysis includes tasks such as detecting and handling null values and identifying outliers within the dataset. In this study, Python libraries such as pandas, NumPy, and Matplotlib were employed for conducting the data analysis.

2.5. Feature selection: -

To reduce the dimensionality of the features of the selected data i.e., genes as we have total 329 samples in the data. In solving equations, the number of (in this case gene profiles) should not exceed the number of equations (the sample size).

Feature selection involves selecting a subset of the original features from the dataset while discarding the irrelevant or redundant ones. The objective is to identify the most informative and discriminate features that contribute significantly to the predictive power of the model. By Eliminating irrelevant features, feature selection simplifies the model and reduces the risk of overfitting, leading to improved performance, interpretability, and reduced computation time.

Scikit-learn has simplified the process of feature selection for us. There are numerous approaches available for feature selection, but they can generally be categorized into three main groups.

1. Filter Methods.
2. Wrapper Methods.
3. Embedded Methods.

1. Filter methods:

These methods apply statistical measures to rank features based on their correlation with the target variable, such as

- I. Pearson Correlation.
- II. Chi-squared method.

- I. Pearson Correlation Feature Selection is a method that quantifies the linear relationship between two continuous variables. It calculates the strength and direction of the linear association using the Pearson correlation coefficient. This technique is particularly effective in identifying linear dependencies between features and the target variable. It performs well when the relationship between features and the target variable follows a linear pattern. (Blessie & Karthikeyan, 2012).

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

r	=	correlation coefficient
x	=	values of the x-variable in a sample
$\bar{x}$	=	mean of the values of the x-variable
y	=	values of the y-variable in a sample
$\bar{y}$	=	mean of the values of the y-variable

II. Chi-Squared Feature Selection, on the other hand, is a method used to assess the independence between two categorical variables. It evaluates the statistical significance of the association using the chi-squared statistic. This approach is suitable for identifying dependencies between categorical features and the target variable. Unlike Pearson correlation, it can capture associations that are nonlinear or non-monotonic in nature.(Bachri & Hatta, n.d.,2017).

$$\chi^2 = \frac{\sum O_i - E_i^2}{E_i}$$

χ^2 = Chi-Square value

O_i = Observed values

E_i = Expected values

In summary, these two feature selection methods differ in the types of variables they handle and the relationships they capture. Pearson correlation is designed for continuous variables and linear relationships, while chi-squared is designed for categorical variables and can capture any type of association between variables.

2. Wrapper methods: -

These methods evaluate different subsets of features by training and testing the model on each subset. This approach considers the performance of the model as the criterion for selecting features

I. Recursive Feature Elimination (RFE): -

RFE is a technique used to select features based on the weights assigned to them by an external estimator, such as the coefficients of a linear model. The process involves recursively considering smaller and smaller sets of features. Initially, the estimator is trained using the full set of features, and the importance of each feature is determined using a specific attribute or callable function. Then, the least important features are removed from the current set. This recursive procedure is repeated on the pruned set of features until the desired number of features to be selected is achieved. The objective is to iteratively identify and retain the most relevant features based on their assigned weights.

3. Embedded techniques: -

Embedded methods encompass feature selection within the learning algorithm itself. These techniques select features during the model training process based on their relevance and impact on the overall model performance. By integrating feature selection directly into the learning algorithm, the need for separate feature selection steps is eliminated. However, embedded methods are algorithm-specific, meaning that each algorithm incorporates its own embedded feature selection mechanisms. This algorithm-specific nature allows embedded methods to optimize the learning algorithm's performance by selecting the most informative features for a particular algorithm. (Saeys et al., 2007)

2.6. Model Training & Building: -

Train/test Split: -

The prior process involves dividing a dataset into two subsets: the training dataset and the test dataset. The training dataset is utilized to train the machine learning model, while the test dataset is kept separate and used to evaluate the model's performance. The model is trained on the training dataset by fitting it to the known inputs and outputs. Subsequently, the test dataset is employed to assess the model's predictive capabilities by providing input elements and comparing the model's predictions to the expected values.

The ultimate objective is to estimate how well the model will perform on new, unseen data that was not used during the training phase. This mimics the real-world scenario where the model is trained on available data and then applied to make predictions on future examples where the expected output or target values are unknown.

2.7. Feature scaling & Handling class imbalance:

Feature scaling is critical in machine learning algorithms involving distance calculations between data points. It ensures that features with higher value ranges do not overpower the calculations, thus maintaining algorithm performance. Machine learning models are sensitive to the relative scales of features, focusing on numeric values rather than ranks. Normalization is crucial for objective functions relying on distance calculations, preventing bias caused by features with wider value ranges. Even without explicit conditions, rescaling features may still be necessary to meet algorithm requirements or avoid saturation phenomena.

Standard Scaler:

The Standard Scaler is designed to work with features that follow a normal distribution. It scales the features individually, ensuring that their distributions are centred around 0 and have a standard deviation of 1. The process of centring and scaling is performed separately for each feature, and the relevant statistics are calculated based on the samples in the training set.

$$x_{stand} = \frac{x_i - u}{\sigma}$$

u = mean of the variable

σ = standard deviation of the variable

Class Imbalance:

Imbalanced classification refers to the issue of classifying data when there is an unequal distribution of classes in the training dataset. This poses a challenge for predictive modelling because many machine learning algorithms are built on the assumption of an equal number of examples for each class. As a result, models trained on imbalanced datasets often exhibit poor predictive performance, especially for the minority class. This is problematic because the minority class is typically of greater importance, making classification errors for this class more significant than for the majority class.

Imblearn Library:

The Imblearn library is specifically developed to address imbalanced datasets by offering a range of techniques such as undersampling, oversampling, and SMOTE (Synthetic Minority Oversampling Technique). SMOTE, a popular oversampling technique, focuses on the minority class by

selecting data points in the feature space and generating new points along a line drawn between them. This process effectively synthesizes additional data points for the minority class, helping to rebalance the dataset. SMOTE is considered one of the most effective methods for oversampling and is widely utilized in handling imbalanced datasets.

Selection of classifier:

We have worked on different kind of classifiers to find out which model works on finding the hidden patterns in the data. Kernel based models like SVM (support vector machines), Logistic Regression (LR) and Tree based models Decision tree (DT), Random Forest (RF) are used.

2.8. Model validation: -

Hyperparameter tuning, also known as hyperparameter optimization, plays a crucial role in the machine learning process. Selecting appropriate hyperparameters can greatly contribute to a model's success in achieving desired performance metrics. Conversely, choosing improper hyperparameters can result in an endless cycle of continuous training and optimization without reaching the desired goals.

Cross-validation is a valuable technique used to assess the effectiveness of a model by training it on a subset of the input data and then evaluating its performance on a separate subset that has not been previously seen. It serves to evaluate how well a statistical model can generalize to an independent dataset. The purpose of cross-validation is to ensure the stability and reliability of the model. Simply relying on the training dataset is insufficient for model fitting. This entire process falls under the umbrella of cross-validation and is distinct from the conventional train-test split approach. (Refaeilzadeh et al., 2009)

Cross-validation overcomes the limitations of the train/test split method by dividing the dataset into groups of train/test splits and averaging the

results. It is particularly useful when optimizing a model's performance that has been trained on the training dataset. Compared to the train/test split, cross-validation offers enhanced efficiency as every observation in the dataset is utilized for both training and testing. This technique finds application in various scenarios, including the comparison of different predictive modelling methods, its relevance in the field of medical research, and its utility in meta-analysis, where data scientists in medical statistics have already embraced its usage.

2.9. Model testing: -

The concept behind constructing machine learning models revolves around a feedback-driven approach. The process involves building a model, receiving feedback through metrics, making necessary improvements, and iterating until a desired level of classification accuracy is attained.

Evaluation metrics play a crucial role in assessing the model's performance. A key attribute of evaluation metrics lies in their ability to differentiate and discern among different model outcomes. Here, we evaluate our model using metrics like Confusion Matrix, Balanced class Accuracy, Class prediction error, ROC_AUC score, Precision, recall, F1-score.

Confusion matrix is a square matrix, typically $n \times n$, where n represents the number of predicted classes. In the specific context discussed here, we have $n = 3$, resulting in a 3×3 matrix. This matrix serves as a performance evaluation tool for classification problems in machine learning when there are multiple classes in the output. It provides a tabular representation of the four different combinations of predicted and actual values.

In order to assess the performance of a model using various metrics, it is common to utilize true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). These metrics are derived by comparing the predicted values from the algorithm with the true values, allowing for the calculation of performance indicators based on these comparative outcomes.

Iterative steps: -

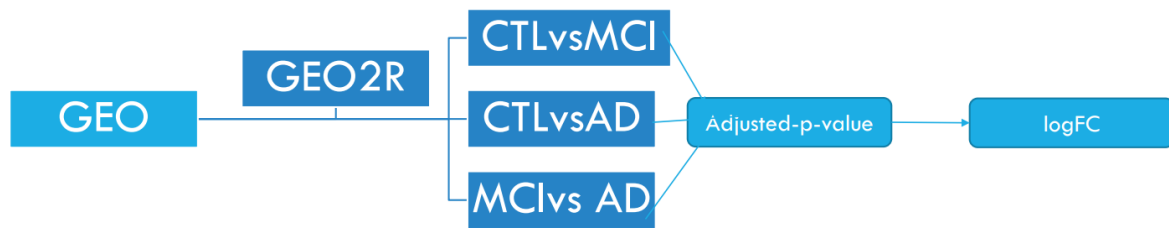
As, we mentioned in fig_2 we have used different feature selection techniques i.e., genes. Here, we explained the iterative steps in two trails as process of experimentation.

Trail_1: -

For Trail_1, we have specifically chosen 27 genes as features, which have been extracted from the research conducted by Soleimani Zakeri et.al. These genes were identified and studied in the same dataset that we are using for our own study. The selected genes include GOLGA1, HLA-B, MBOAT1, RABL2B, ARMC7, IL10RB, STX5, TFIP11, VCP, ALDOC, APBA3, CHST14, DDX19A, HNRNPUL1, KCTD2, LAMTOR1, RPA1, PLAGL2, CREBRF, LCOR, ALDOA, LPP, KLF13, CANX, MRI1, STX16, and SLC38A1. These genes have been identified as Differentially Expressed Genes (DEGs) in the aforementioned study..(Soleimani Zakeri et al., 2020)

Trail_2: -

In Trail_2, First the selected dataset was analysed using GEO2R which is an inbuilt tool in GEO database which works on 'LIMMA' library one of the packages of R Bioconductor programme. The samples are tagged with CTL, MCI & AD respectively and the Gene expression analysis was done for three times for CTL vs MCI, CTL vs AD, MCI vs AD. Second, the related IDs were assigned to the genes, and any duplicated or genes without IDs were removed. The genes were then arranged based on their adjusted p-value, and the genes with a significant adjusted p-value of less than 0.05 were chosen for additional analysis, while the rest were disregarded.

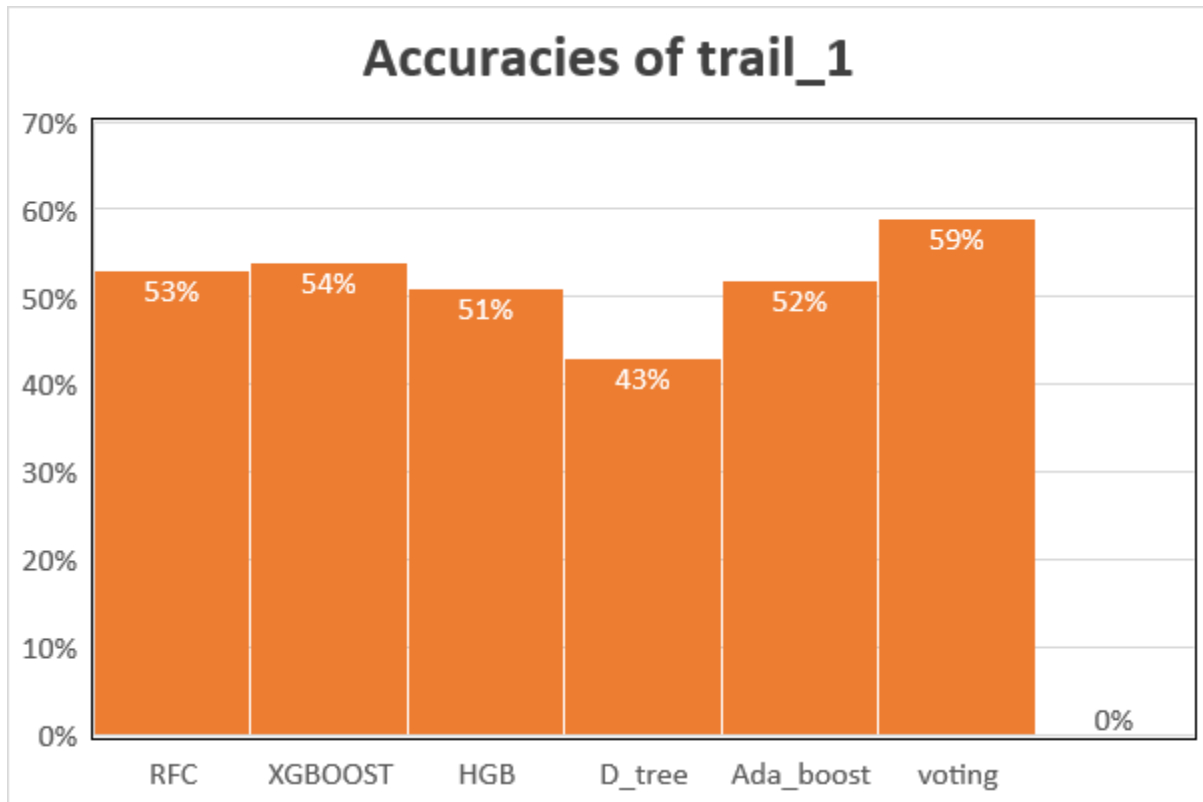


Fig_3: Extraction of genes from GEO using GEO2R tool.

Then from the remaining genes according to the LogFc values top 200 genes which are upregulated and downregulated respectively from Each analysis. Total of 700 genes were selected for further analysis. Then different feature selection techniques are used for selection of important genes to reduce the dimensional space of the features. The feature size was put down 200 so, from different feature selection techniques different genes are selected and selected genes were used as input to the model.

3.RESULTS

3.1. Trail_1 result: -



Fig_4: Comparison graph of accuracies of Different classifiers.

After choosing the genes derived from the research conducted by Soleimani Zakeri, the provided illustration demonstrates the effectiveness of various machine learning models in the experiment. When compared to the decision tree classifier, the remaining classifiers exhibit higher accuracy, exceeding 50%. The decision tree classifier specifically achieves an accuracy of 43%. On the other hand, the voting classifier outperforms all other models with the highest accuracy of 59%. The voting classifier is a composite model trained on multiple base models, such as RFC, HGB, XGBoost, and Adaboost, which are considered as the major performing classifiers in this scenario.

3.2. Trail_2 results: -

We have used different feature selection techniques in trail_2 as mentioned and selected features were used as input to algorithms. Here, we used specific classifiers like SVC & RFC as these are less prone to overfitting and works efficiently on small size datasets.

Training on Pearson selected features: -

The Pearson selected features are trained on RFC & SVC and the results obtained are visualized in following figures.

1. RFC with Pearson selected feature

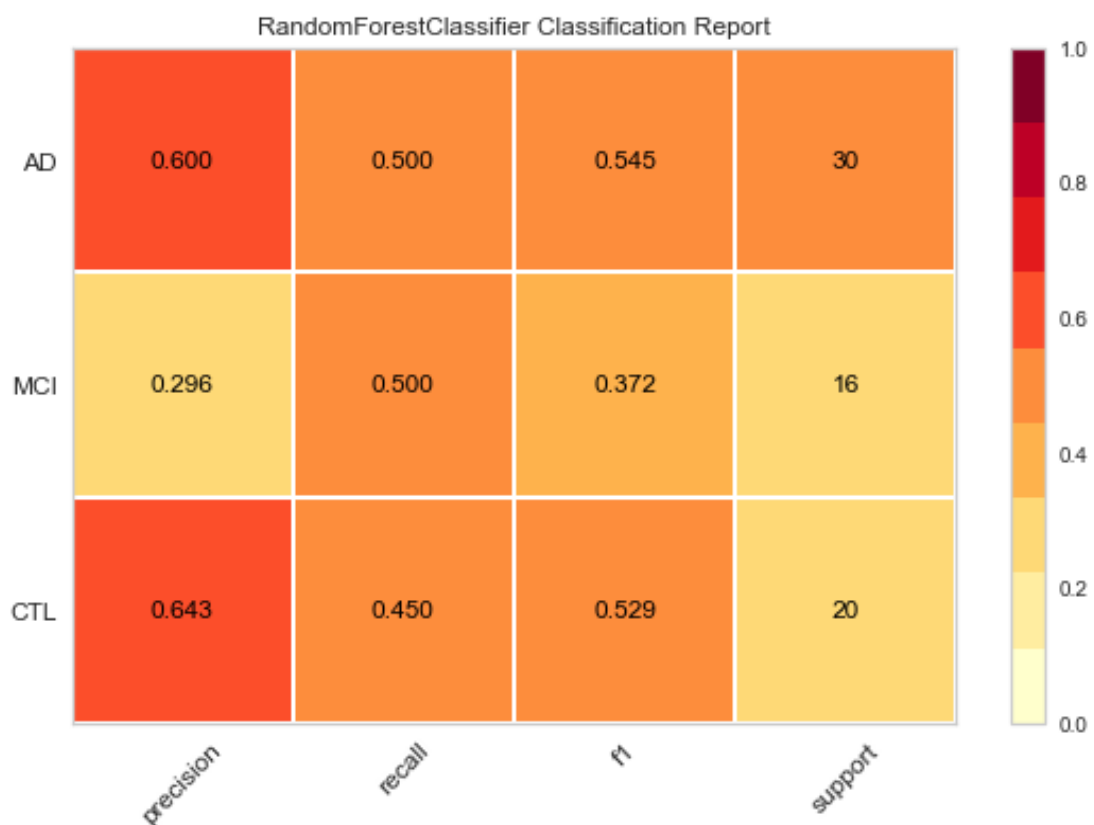


Fig 5.1: classification report heat map of Pearson method selected genes using RFC.

The precision and recall values are higher for the CTL class, indicating better performance compared to the MCI and AD classes. The MCI class presents lower precision and recall, indicating more difficult predictions.

However, the AD class demonstrates moderate precision and recall, suggesting relatively better prediction performance compared to MCI.

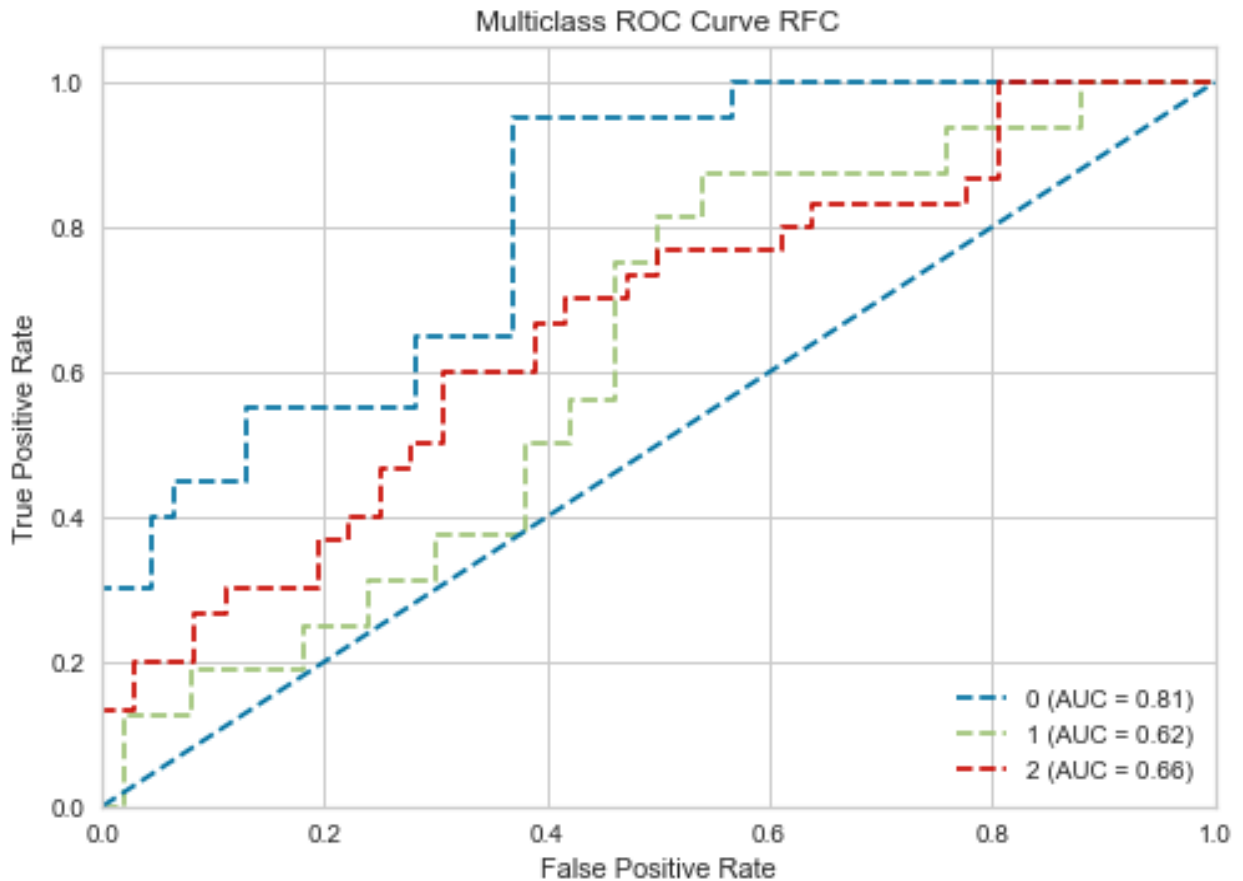


Fig 5.2: ROC curve of Pearson method selected genes using RFC

The CTL class achieves a relatively high ROC-AUC score of 0.81, indicating good discrimination and a high probability of correct classification. The MCI class has a lower ROC-AUC score of 0.62, suggesting a moderate ability to distinguish between MCI and other classes. The AD class obtains a ROC-AUC score of 0.66, indicating a reasonable performance in classifying AD cases. Overall, the model shows better discrimination for the CTL class, while the MCI and AD classes exhibit comparatively lower scores, indicating the need for further improvement in classification accuracy.

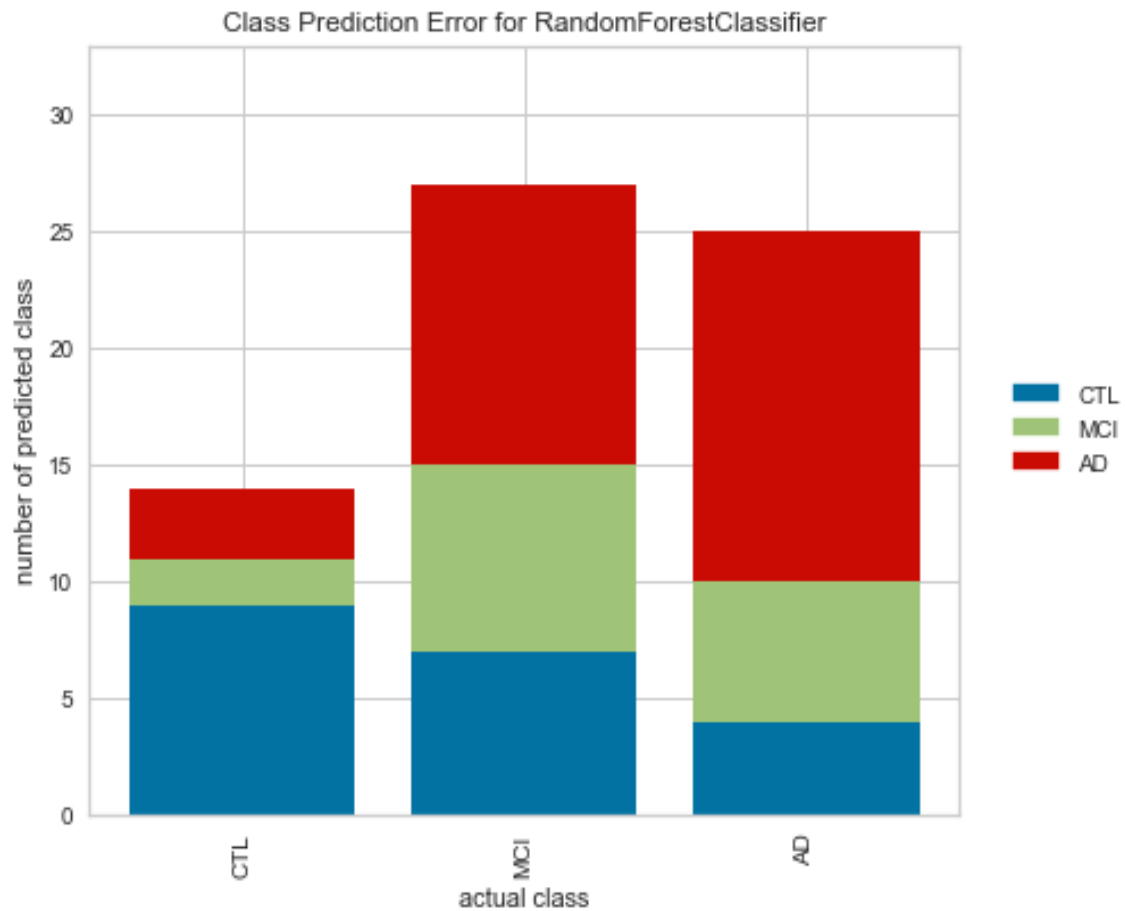


Fig 5.3: class prediction error of Pearson method selected genes using RFC

The RFC model, which trained on Pearson-selected genes, exhibits accurate predictions for the CTL class. However, it encounters challenges when predicting the other classes, resulting in a higher confusion rate. This is evident from the comparison of class prediction error rates, where the CTL class has a lower error rate compared to MCI and AD classes.

2. SVC with Pearson selected features:

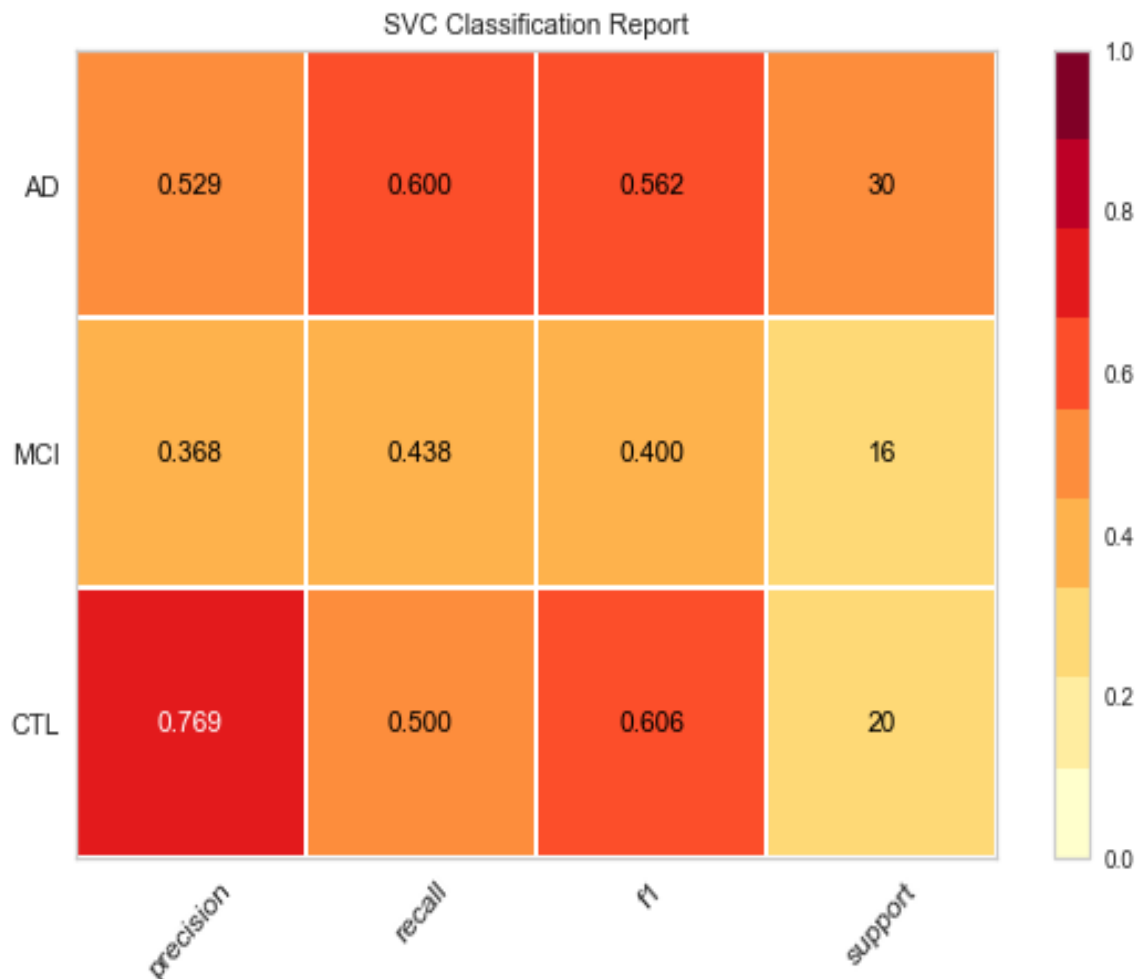
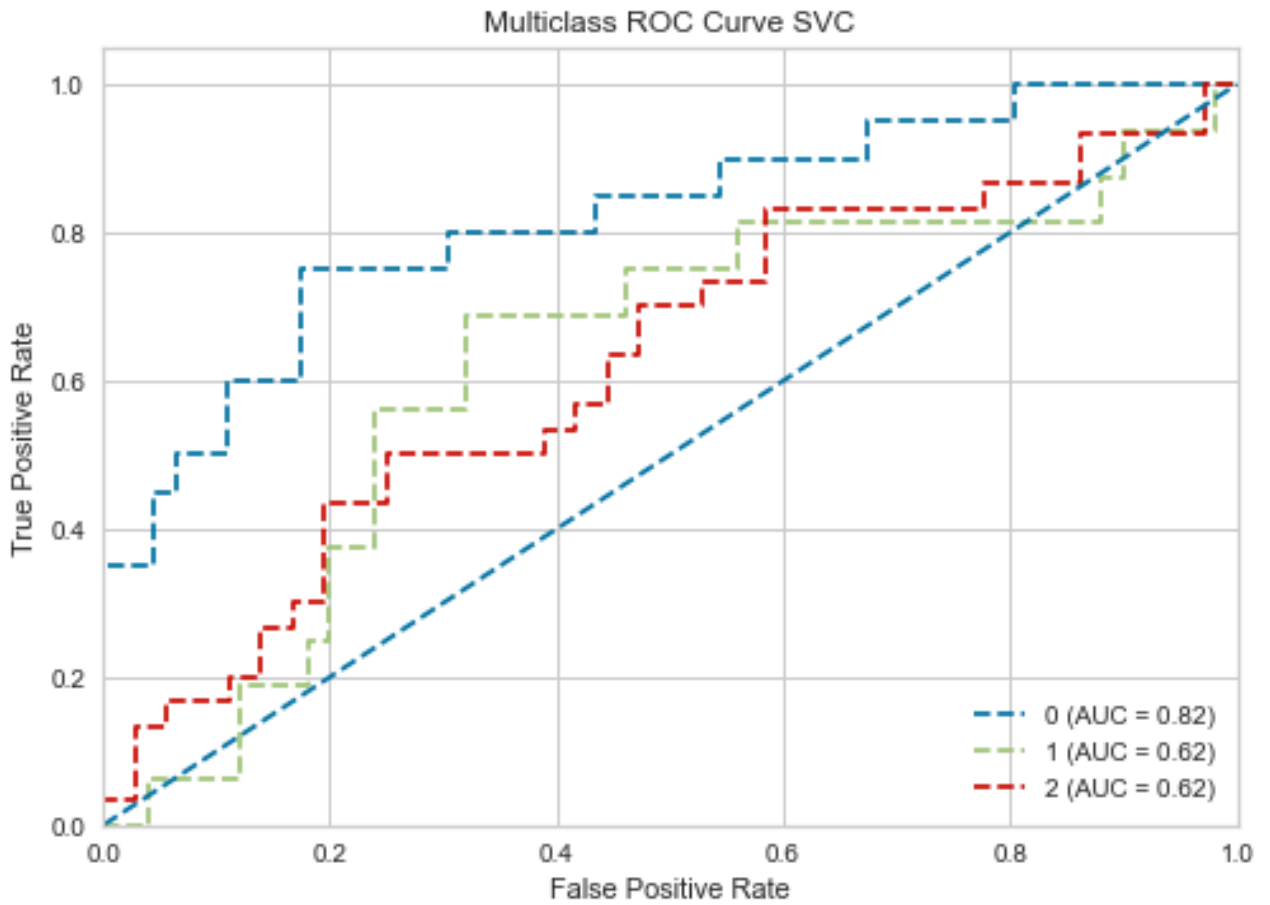


Fig 6.1: classification report heat map of Pearson method selected genes using SVC.

The precision and f1 values are higher for the CTL class, indicating better performance compared to the MCI and AD classes. The MCI class presents lower precision and recall, indicating more difficult predictions. However, the AD class demonstrates moderate precision and recall, suggesting relatively better prediction performance compared to MCI.



Fig_6.2: ROC curve of Pearson method selected genes using SVC.

The ROC-AUC scores provide an overall measure of the model's performance in distinguishing between different classes. In this case, the CTL class achieves a relatively high ROC-AUC score of 0.82, indicating good discrimination ability. On the other hand, the MCI and AD classes have lower scores of 0.62 each, suggesting a moderate performance in class separation. These results suggest that the model performs relatively well in identifying the CTL class compared to the MCI and AD classes.

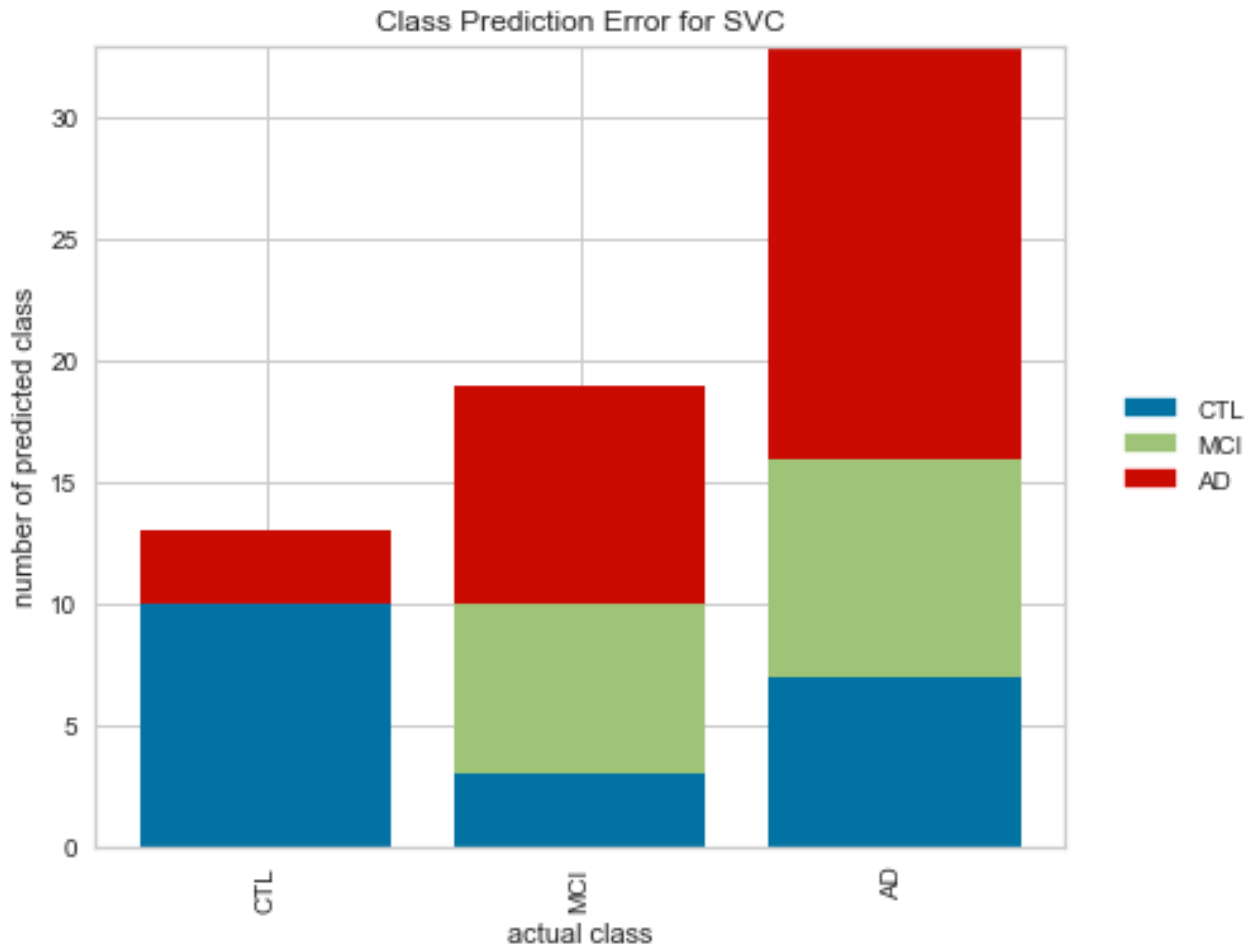


Fig 6.3: Class prediction error of Pearson method selected genes using SVC.

The SVC model, which utilized Pearson-selected genes, exhibits accurate predictions for the CTL class. However, it encounters challenges when predicting the other classes, resulting in a higher confusion rate. This is evident from the comparison of class prediction error rates, where the CTL class has a lower error rate compared to MCI and AD classes.

Training on Chi-square selected features: -

1.RFC with Chi-square features: -

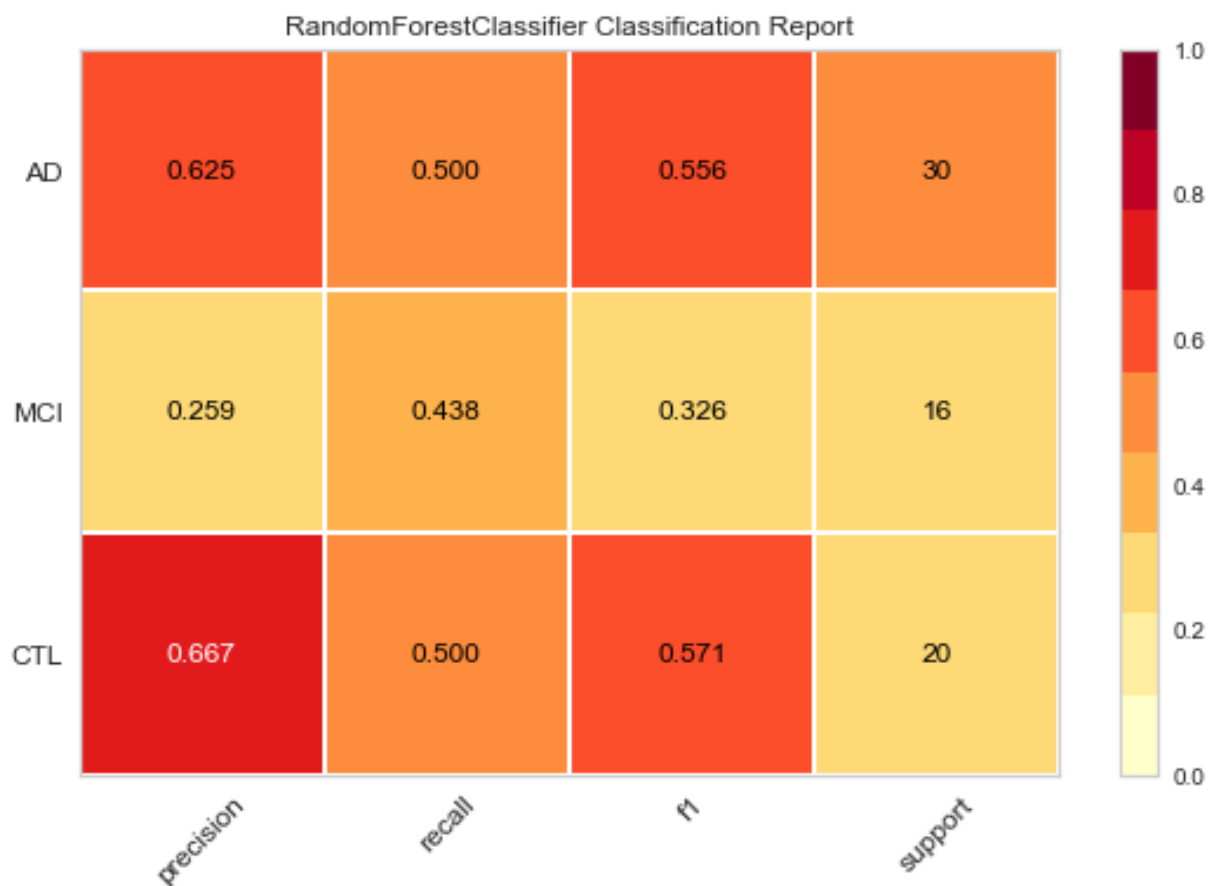
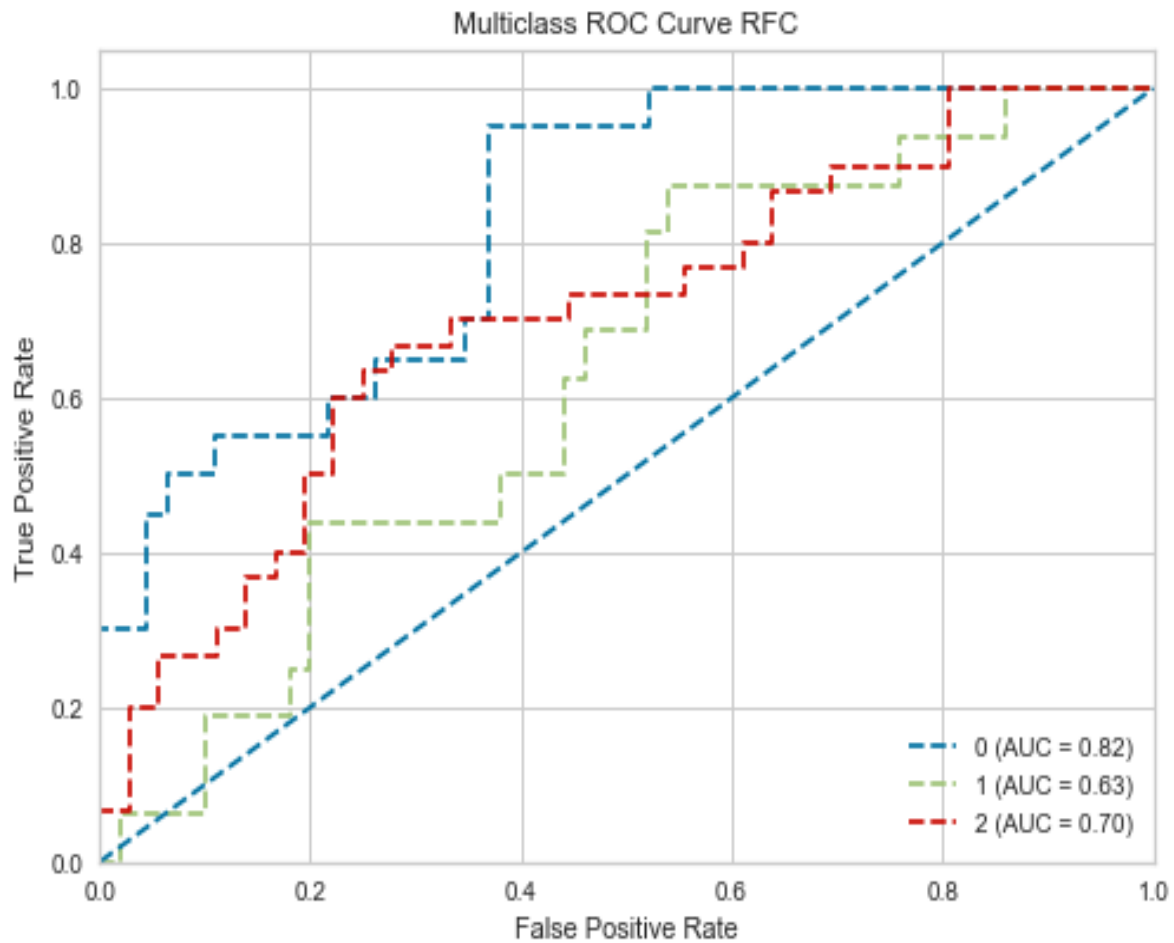


Fig 7.1: classification report heat map of Chi-square method selected genes using RFC.

The precision and f1 values are higher for the CTL class, indicating better performance compared to the MCI and AD classes. The MCI class presents lower precision and recall, indicating more difficult predictions. However, the AD class demonstrates moderate precision and recall, suggesting relatively better prediction performance compared to MCI.



Fig_7.2: ROC curve of Chi-square method selected genes using RFC.

The class CTL achieves a relatively high ROC-AUC score of 0.82, indicating good discrimination ability in distinguishing CTL instances from other classes. The MCI class shows a moderate ROC-AUC score of 0.63, suggesting a somewhat lower performance in distinguishing MCI instances. The AD class achieves a ROC-AUC score of 0.70, indicating a fair ability to differentiate AD instances from other classes. Overall, the model performs relatively well for the CTL class and relatively less effectively for the MCI and AD classes in terms of ROC-AUC scores.

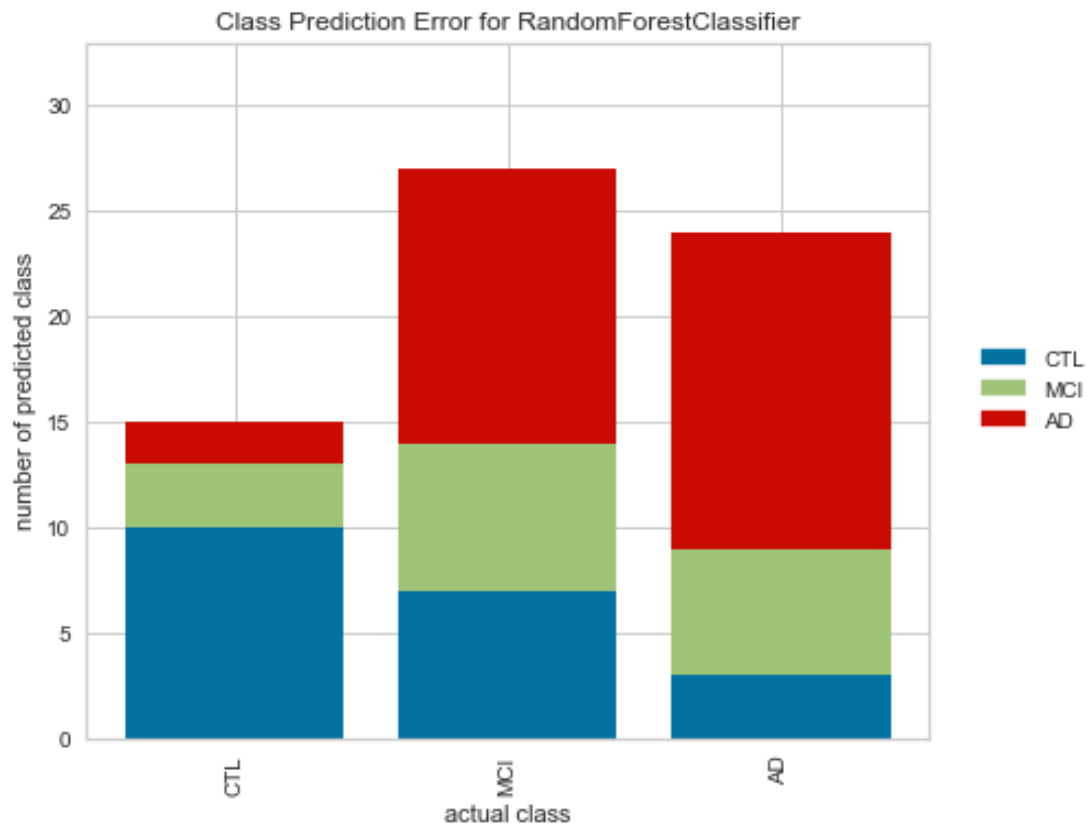


Fig 7.3: class prediction error of Chi-square method selected genes using RFC.

The RFC model, which utilize Chi-square selected genes, exhibits accurate predictions for the CTL class. However, it encounters challenges when predicting the other classes, resulting in a higher confusion rate. This is evident from the comparison of class prediction error rates, where the CTL class has a lower error rate compared to MCI and AD classes.

2. SVC with Chi-square features: -

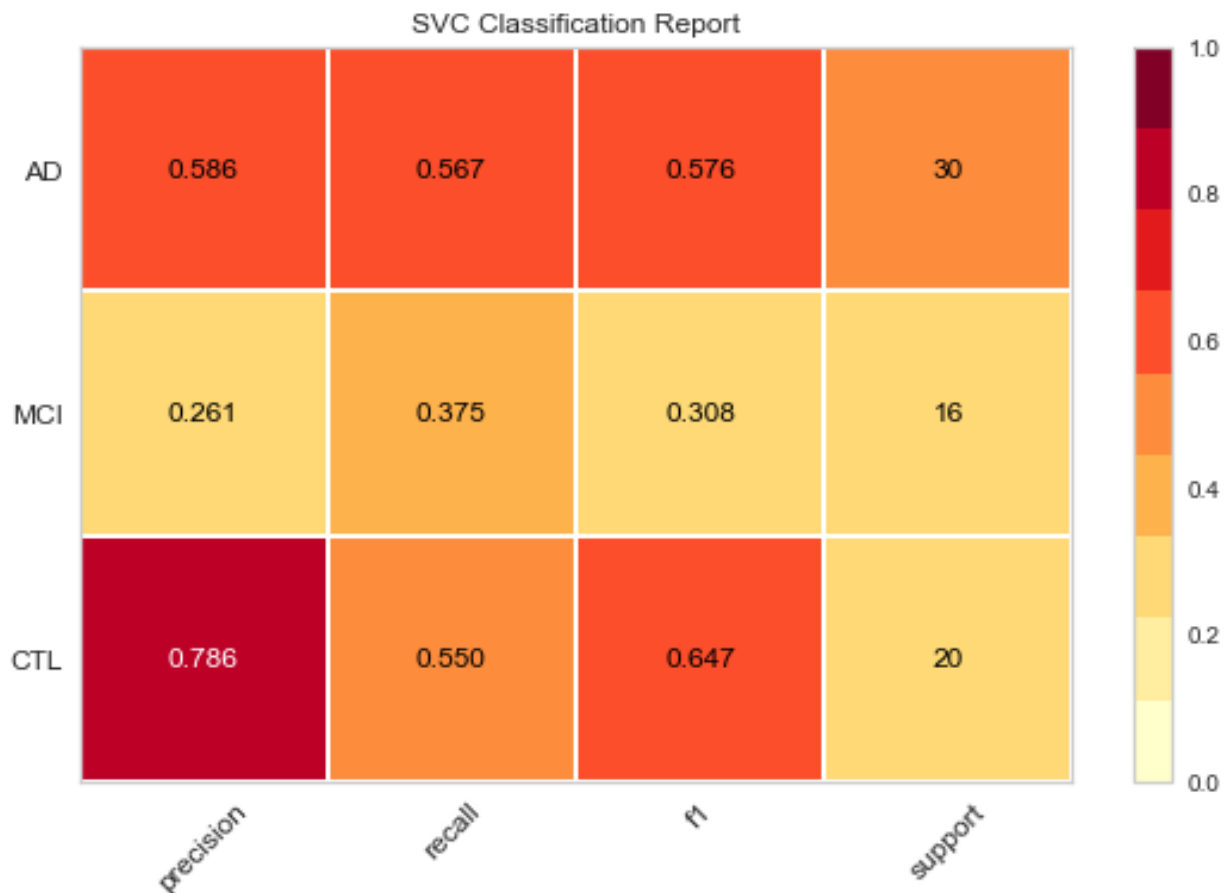


Fig 8.1: classification report heat map of Chi-square method selected genes using SVC.

The precision and f1 values are higher for the CTL class, indicating better performance compared to the MCI and AD classes. The MCI class presents lower precision and recall, indicating more difficult predictions. However, the AD class demonstrates moderate precision and recall, suggesting relatively better prediction performance compared to MCI.

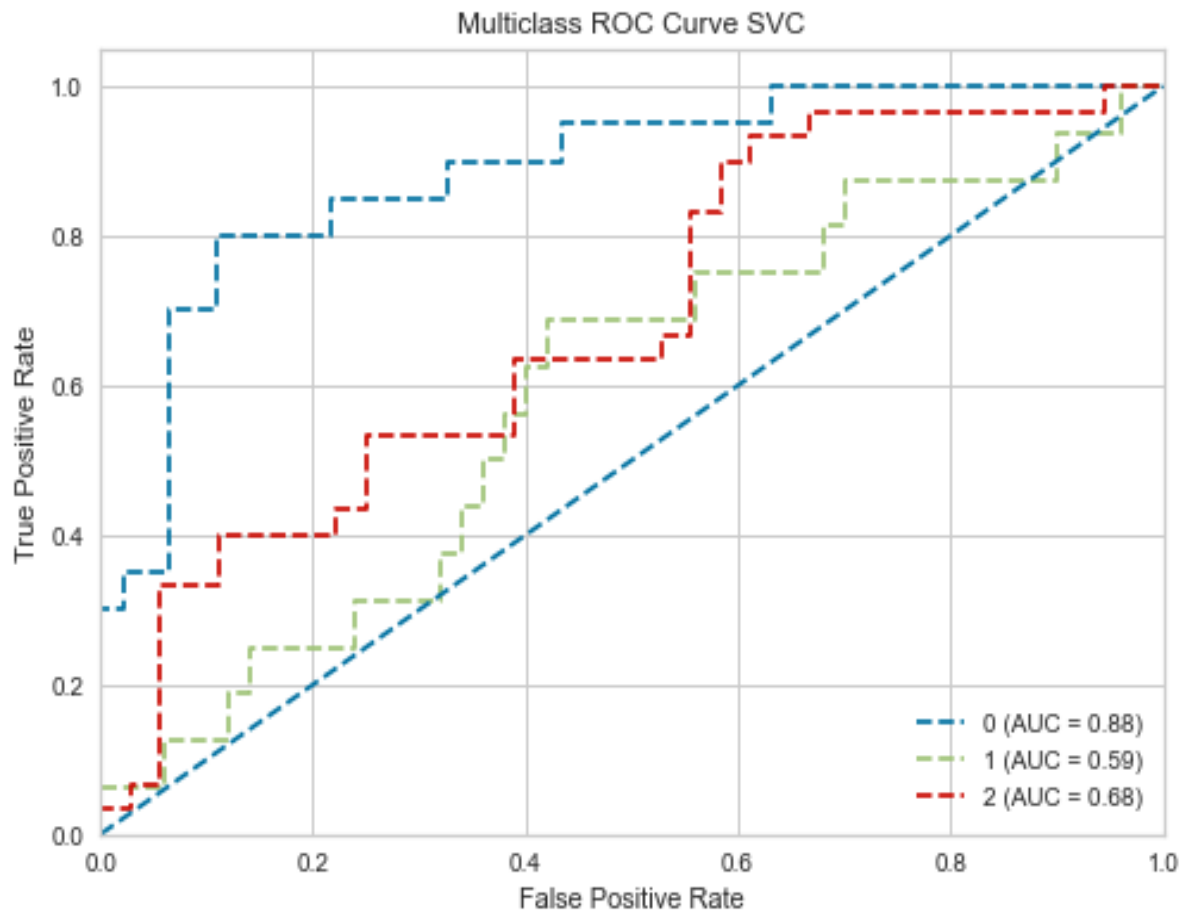


Fig 8.2: ROC curve of Chi-square method selected genes using SVC.

The class CTL achieves a relatively high score of 0.88, suggesting that the model performs well in correctly classifying instances belonging to this class. The MCI class has a lower score of 0.59, indicating a moderate performance in distinguishing MCI instances from the other classes. The AD class achieves a score of 0.68, suggesting a relatively better performance compared to MCI but still with room for improvement in accurately identifying instances of AD.

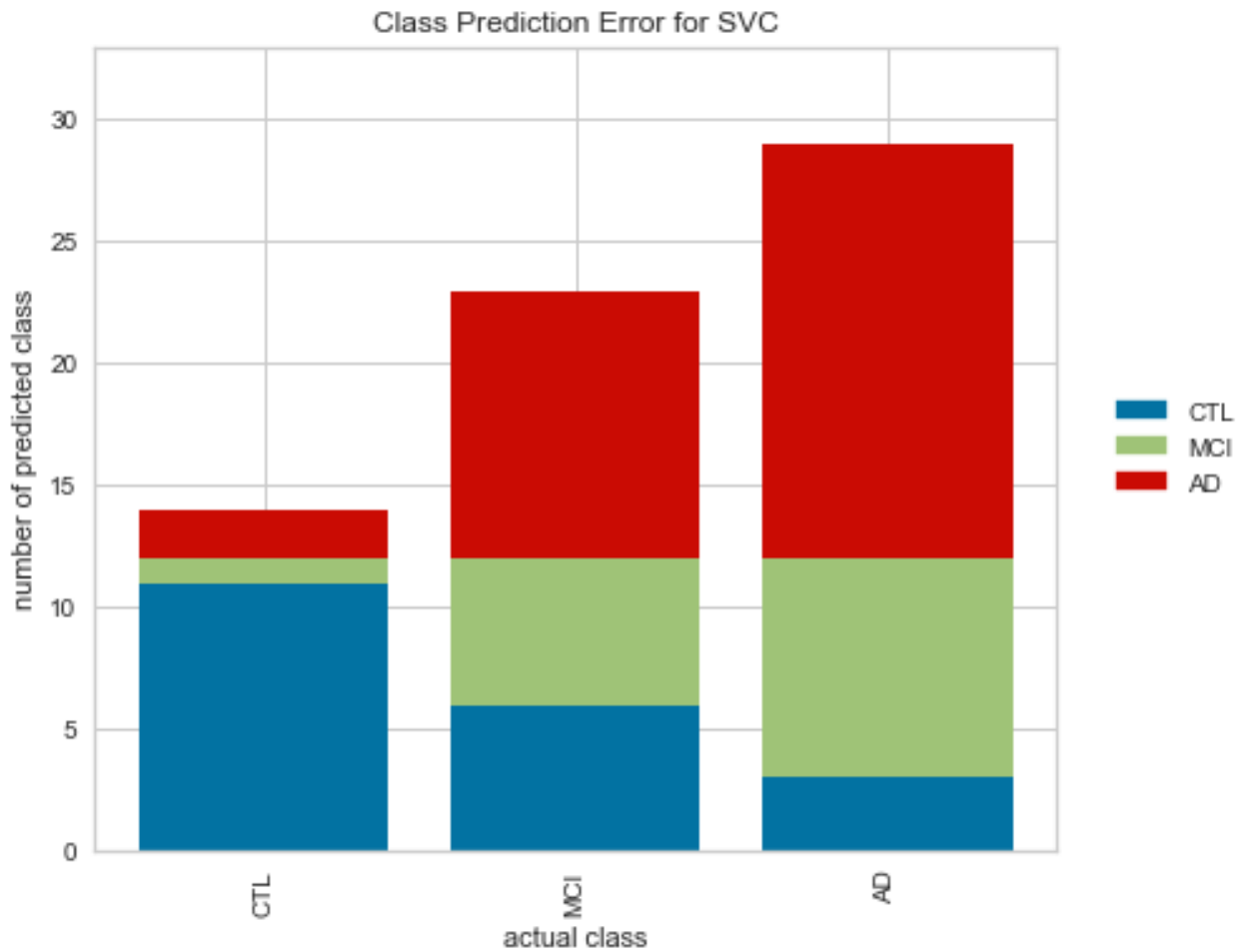


Fig 8.3: class prediction error of Chi-square method selected genes using SVC.

The SVC model, which utilize chi-square selected genes, exhibits accurate predictions for the CTL class. However, it encounters challenges when predicting the other classes, resulting in a higher confusion rate.

Training on RFE selected features

1.RFE selected features with RFC:



Fig 9.1: classification report heat map of RFE method selected genes using RFC.

The precision and f1 values are higher for the CTL class, indicating better performance compared to the MCI and AD classes. The MCI class presents lower precision and recall, indicating more difficult predictions. However, the AD class demonstrates moderate precision and recall, suggesting relatively better prediction performance compared to MCI.

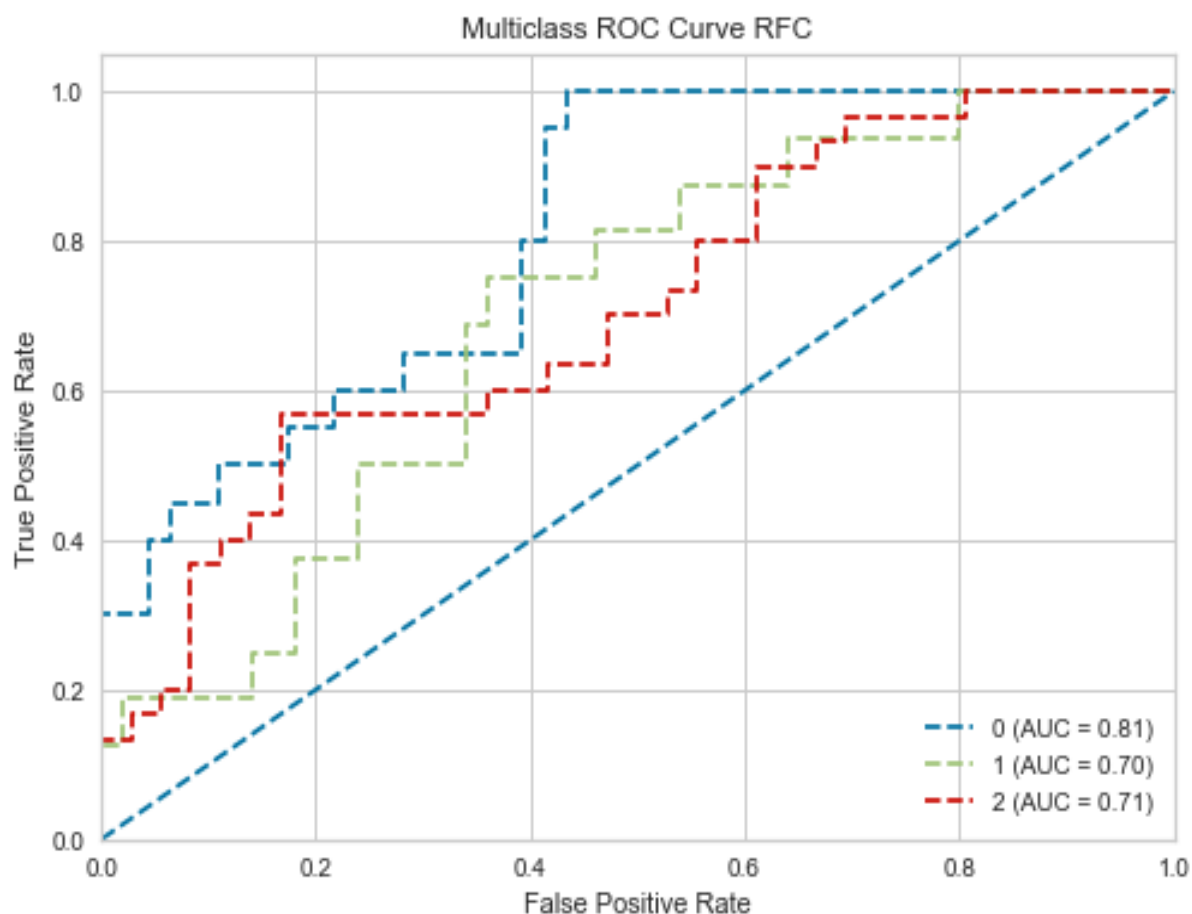


Fig 9.2: ROC curve of RFE method selected genes using RFC.

For the CTL class, the ROC-AUC score of 0.81 suggests that the model demonstrates good discrimination ability in correctly classifying instances belonging to this class. The MCI class has a slightly lower ROC-AUC score of 0.70, indicating a moderate performance in distinguishing MCI instances from other classes. The AD class has a similar performance with a ROC-AUC score of 0.71, suggesting a relatively better ability to differentiate AD instances. Overall, the model shows varying levels of performance across the different classes, with the CTL class performing relatively better and the MCI and AD classes showing moderate discrimination capabilities.

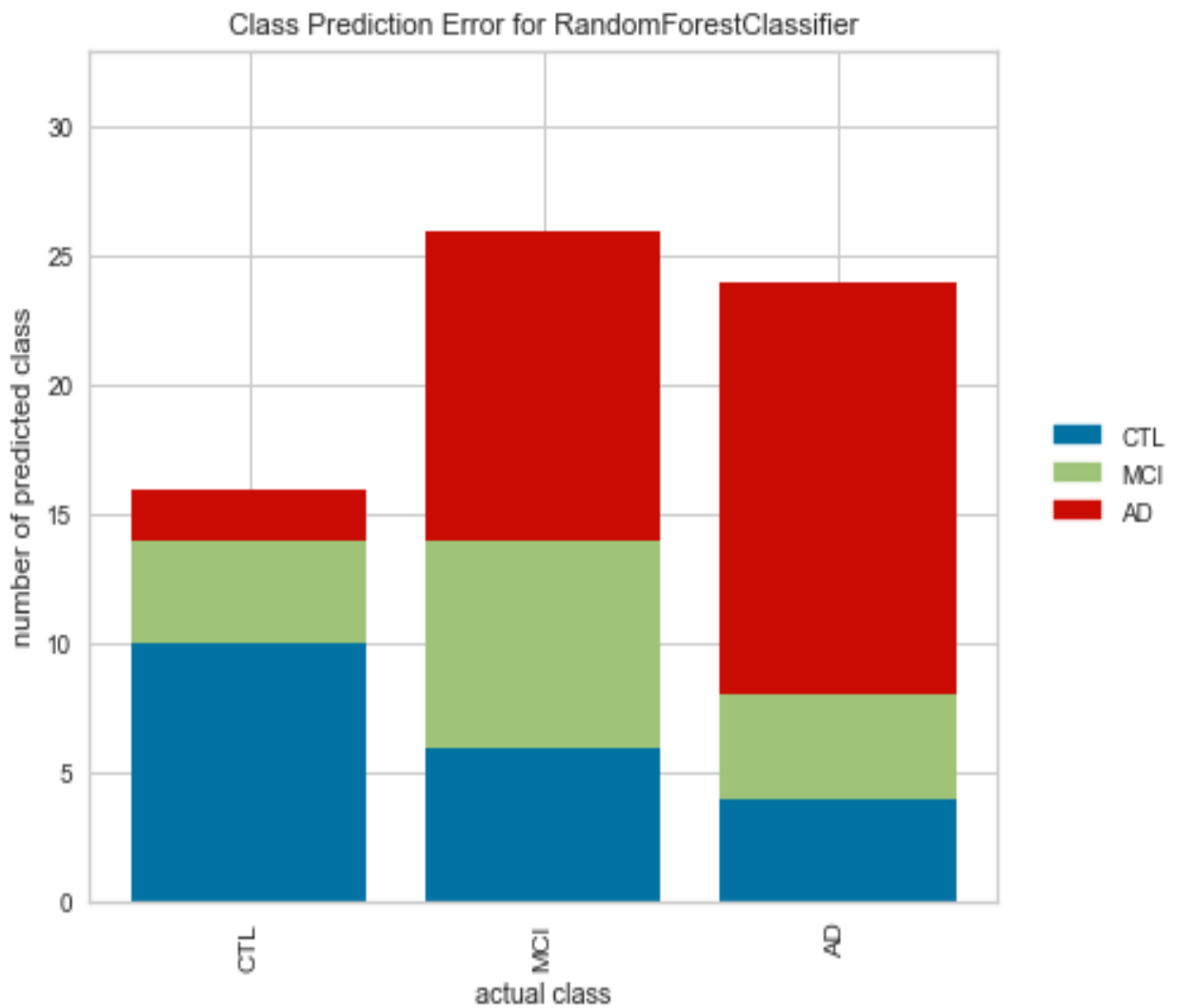
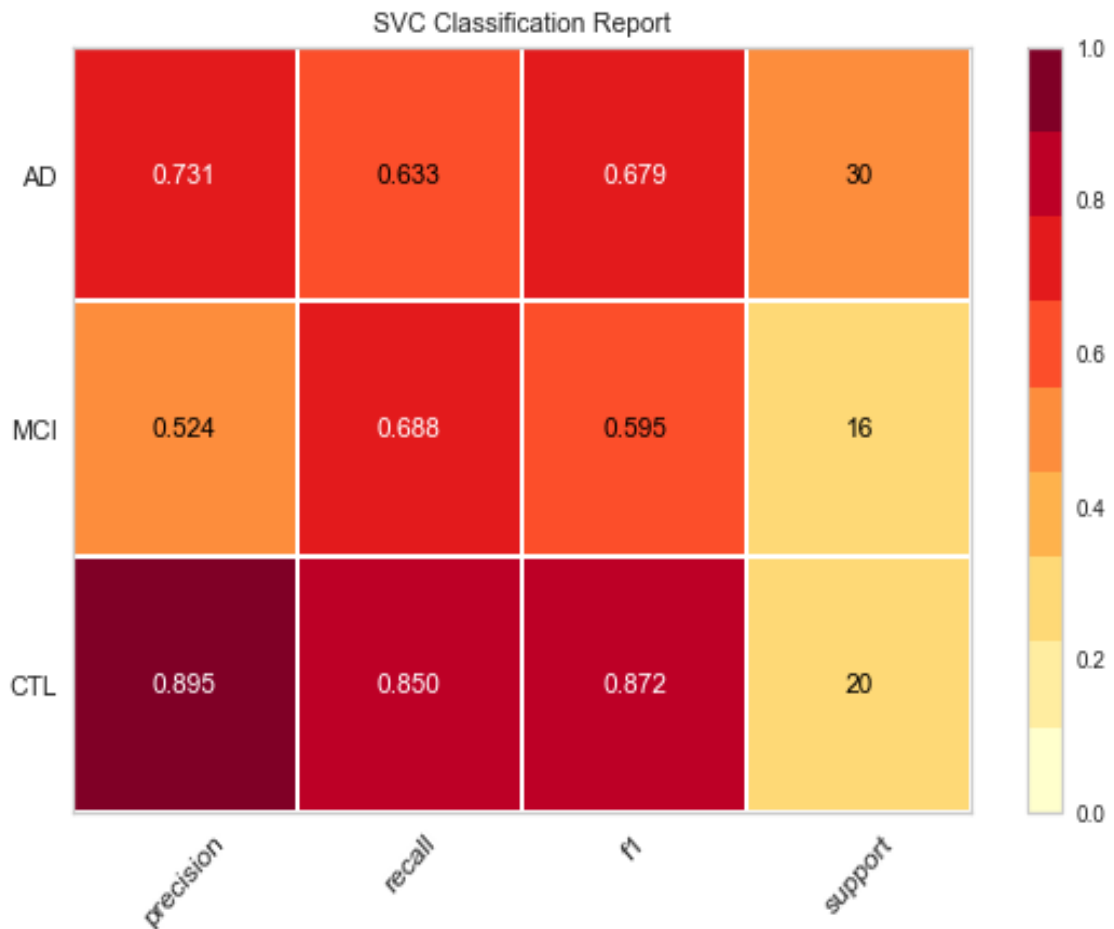


Fig 9.3: class prediction error rate of RFE method selected genes using RFC.

The RFC model, which utilized RFE-selected genes, exhibits accurate predictions for the CTL class. However, it encounters challenges when predicting the other classes, resulting in a higher confusion rate between other classes.

RFE selected features with SVC: -



Fig_10.1: classification report heat map on RFE method selected genes using SVC.

The precision and f1 values are higher for the CTL class, indicating better performance compared to the MCI and AD classes. The MCI class performs very well compared to previous models though the values are quite low compared to CTL & AD.

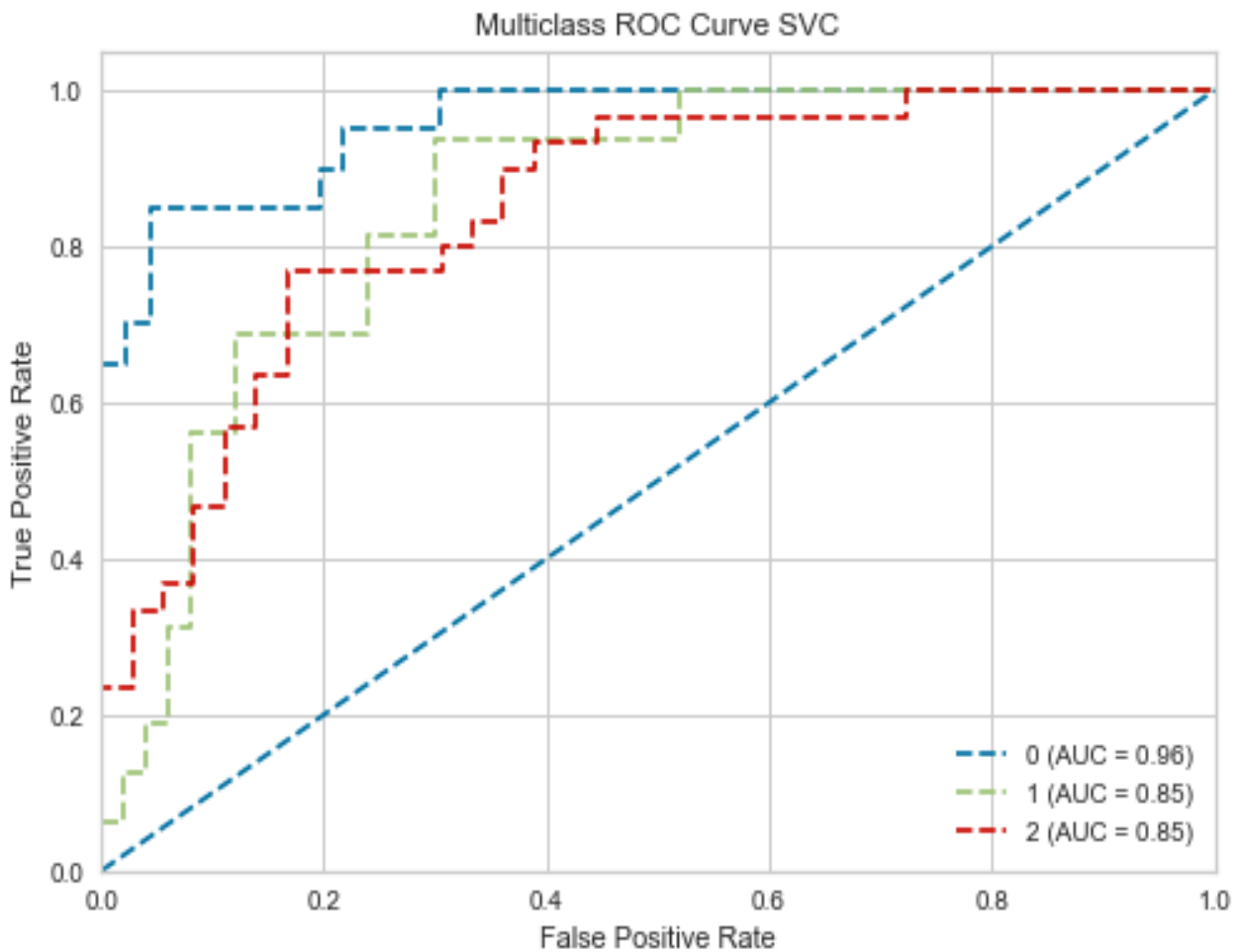
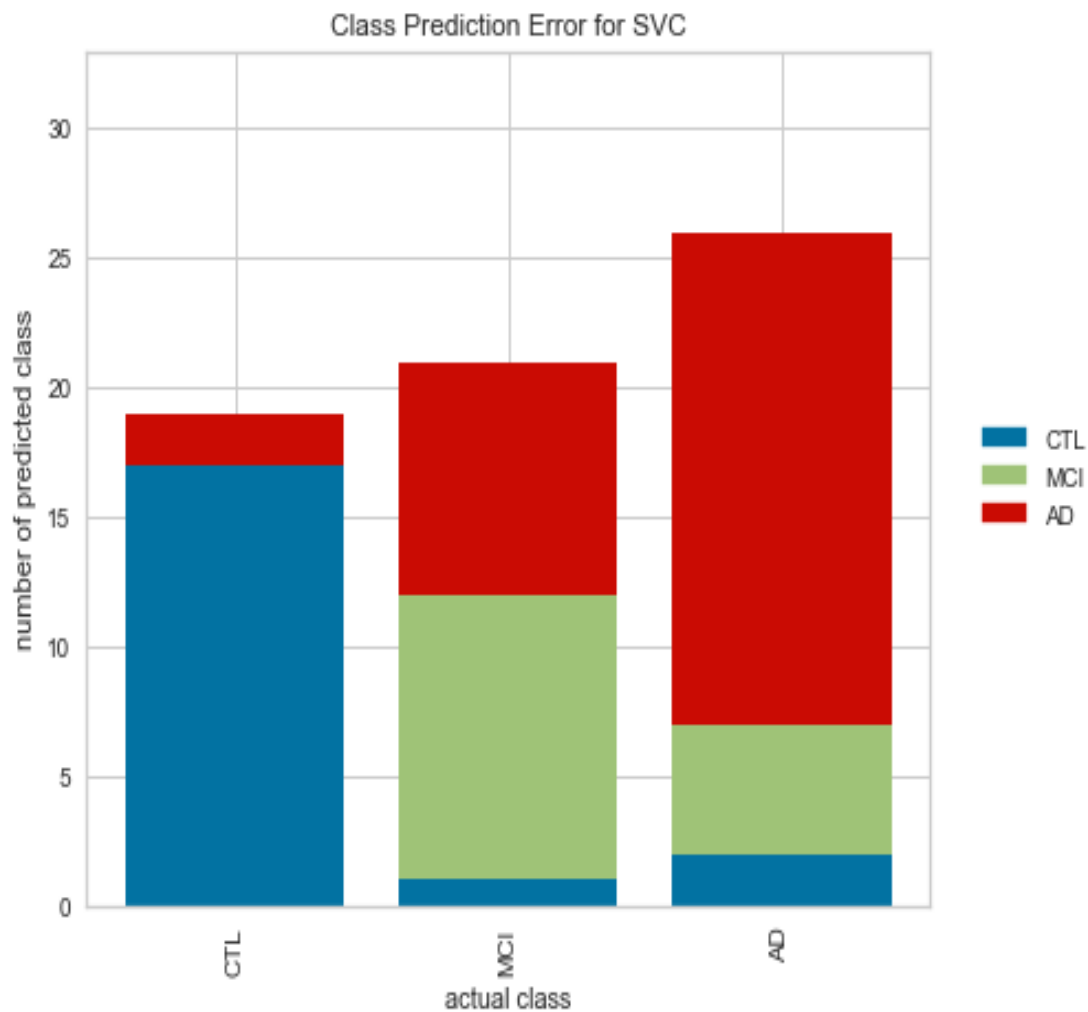


Fig 10.2: ROC curve of RFE method selected genes using SVC.

In this case, the CTL class achieves a high ROC-AUC score of 0.96, indicating excellent discrimination and a strong ability to correctly classify instances of the CTL class. The MCI and AD classes also demonstrate relatively good performance with ROC-AUC scores of 0.85 each, suggesting a moderate ability to distinguish between instances of these classes. Overall, the model shows promising performance in classifying instances across the CTL, MCI, and AD classes, with the CTL class exhibiting the highest discriminatory power.

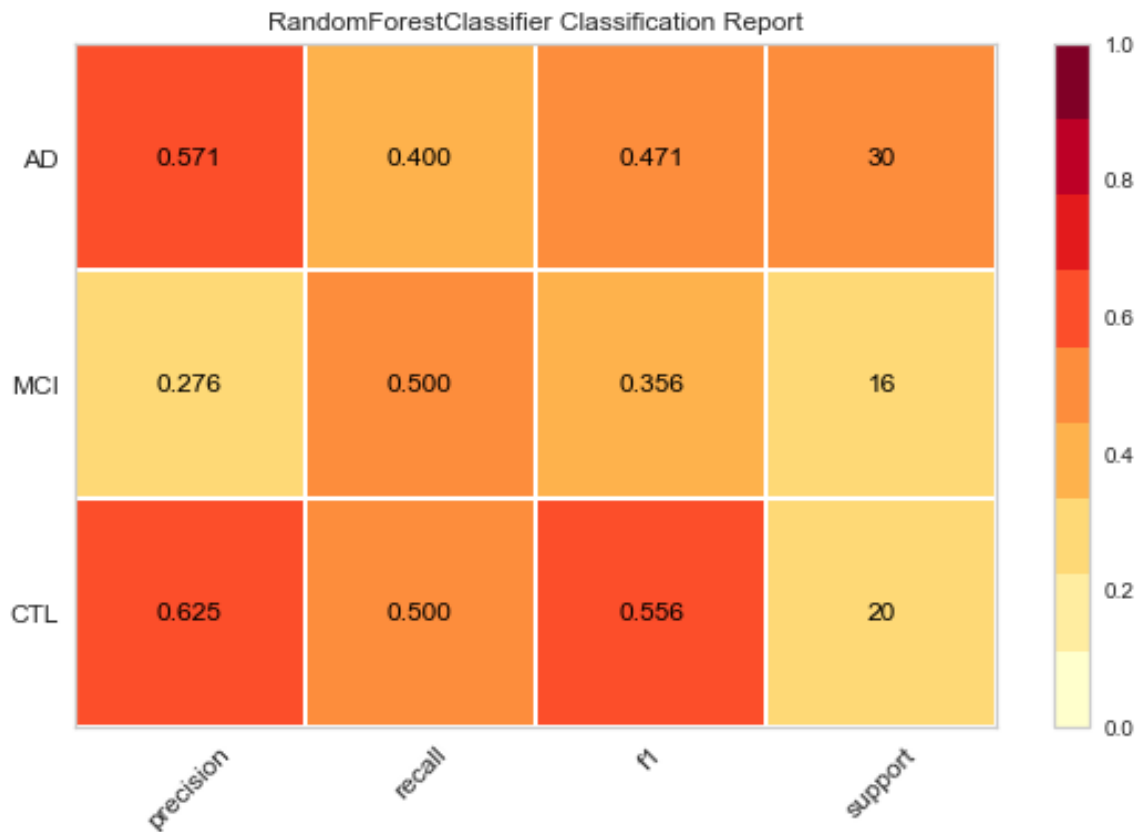


Fig_10.3: Class prediction error rate of RFE method selected genes using SVC

The SVC model, utilizing RFE-selected genes, demonstrates precise predictions for all classes compared to other models, resulting in a lower confusion rate. However, distinguishing the decision boundary between the MCI and AD classes still requires additional research for better clarity.

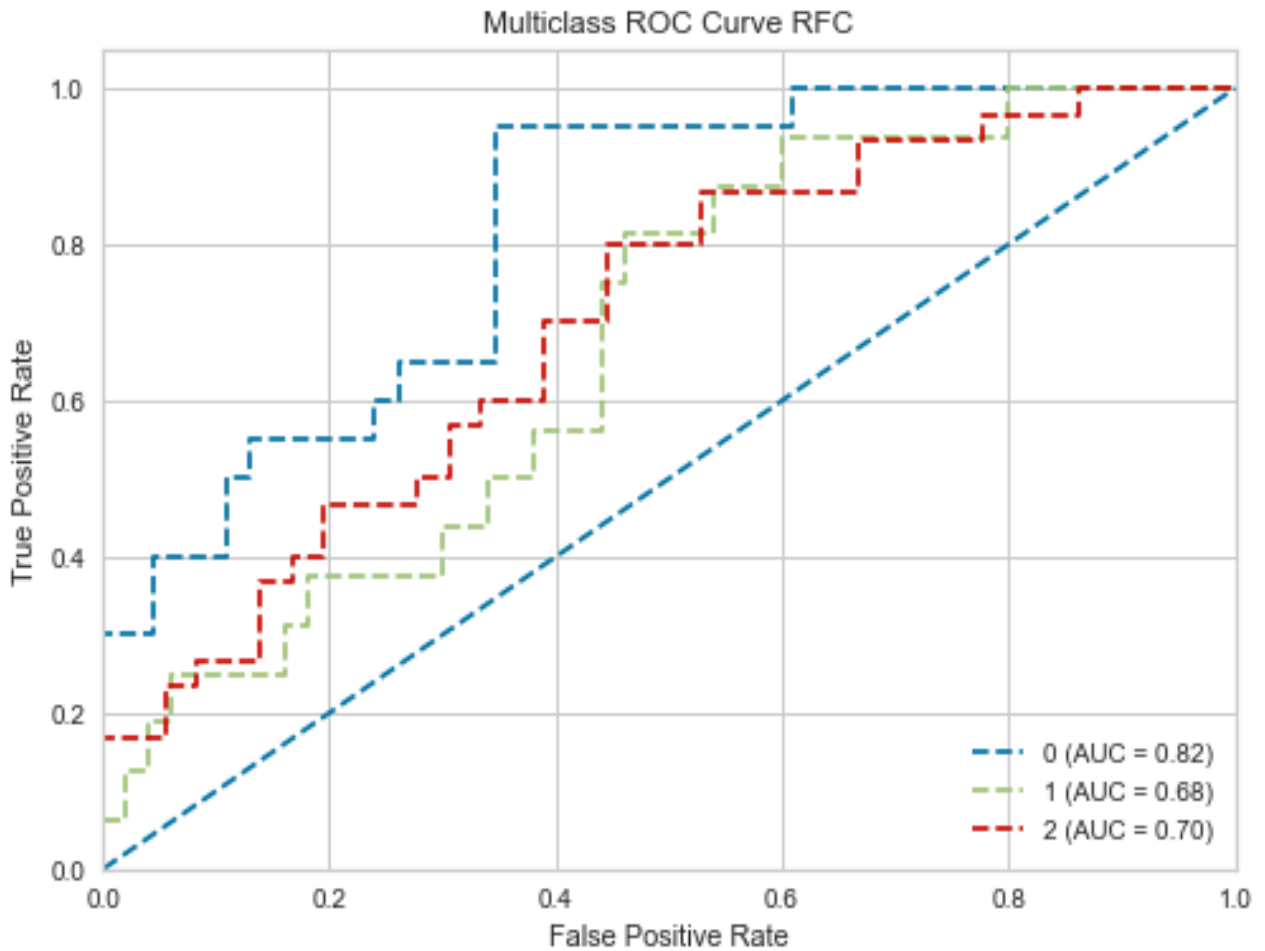
Training on Embedded method selected features:

Embedded selected features with RFC estimator:



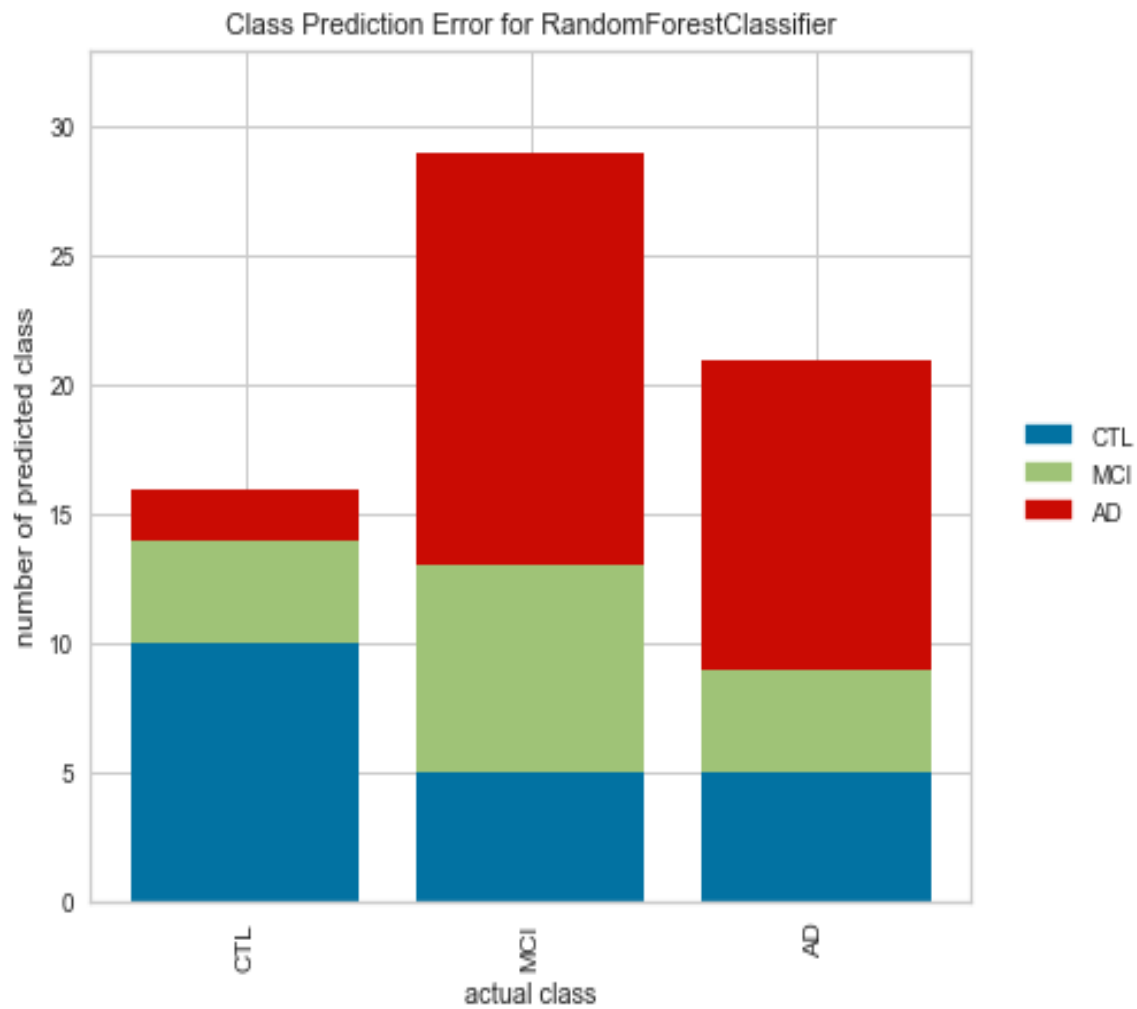
Fig_11.1: Classification report of Embedded method selected genes using RFC

The precision and f1 values are higher for the CTL class, indicating better performance compared to the MCI and AD classes. The MCI class presents lower precision and recall, indicating more difficult predictions. However, the AD class demonstrates moderate precision and recall, suggesting relatively better prediction performance compared to MCI.



Fig_11.2: ROC curve of Embedded method selected genes using RFC.

The class CTL achieves a relatively high ROC-AUC score of 0.82, suggesting that the model is able to effectively distinguish CTL instances from other classes. The MCI class has a lower ROC-AUC score of 0.68, indicating a somewhat moderate ability to discriminate MCI instances. The AD class performs similarly to the MCI class, with a ROC-AUC score of 0.70, suggesting a moderate discriminatory performance. Overall, the model shows better discrimination for the CTL class compared to the MCI and AD classes based on the ROC-AUC scores.



Fig_11.3: Class prediction error rate of Embedded method selected genes using RFC

The RFC model, which utilized Embedded method selected genes, exhibits accurate predictions for the CTL class. However, it encounters challenges when predicting the other classes, resulting in a higher confusion rate. This is evident from the comparison of class prediction error rates, where the CTL class has a lower error rate compared to MCI and AD classes.

Embedded features with SVC:

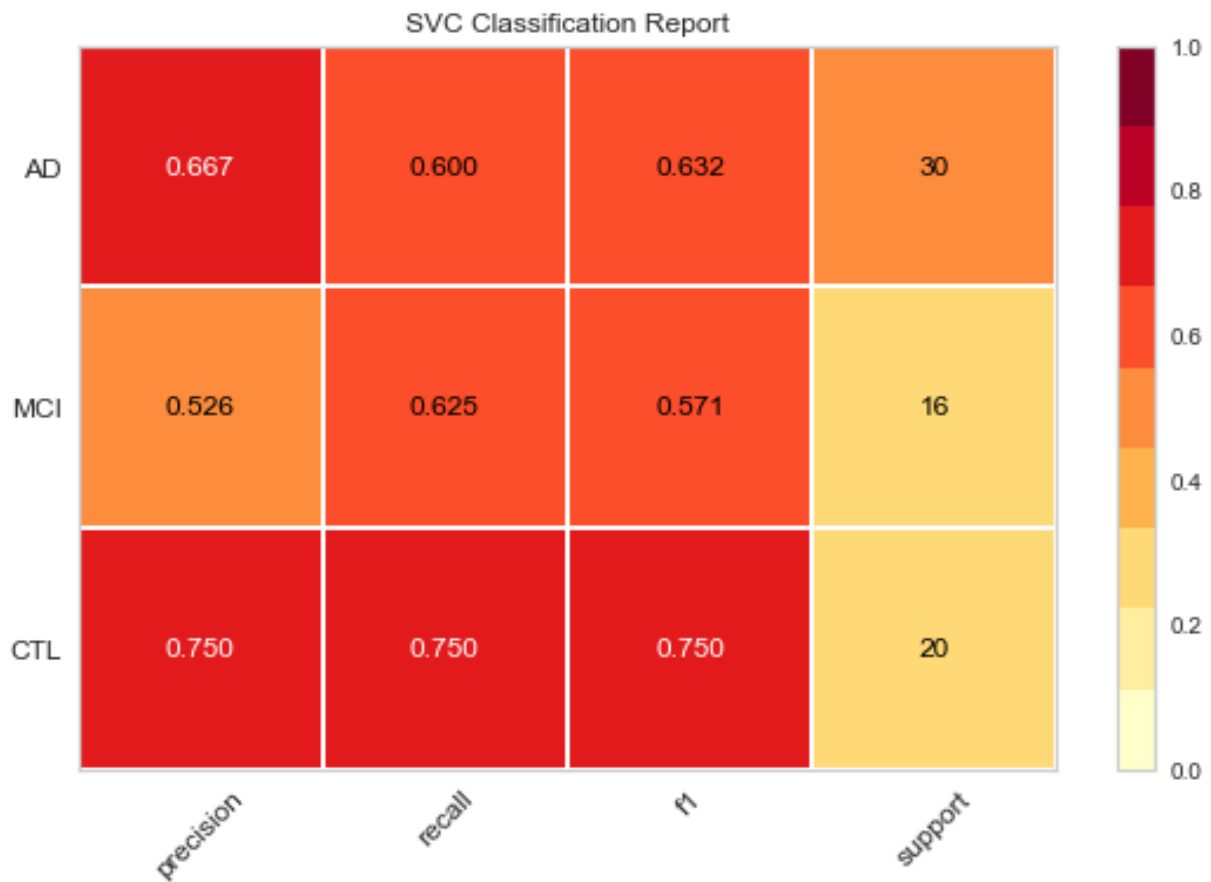
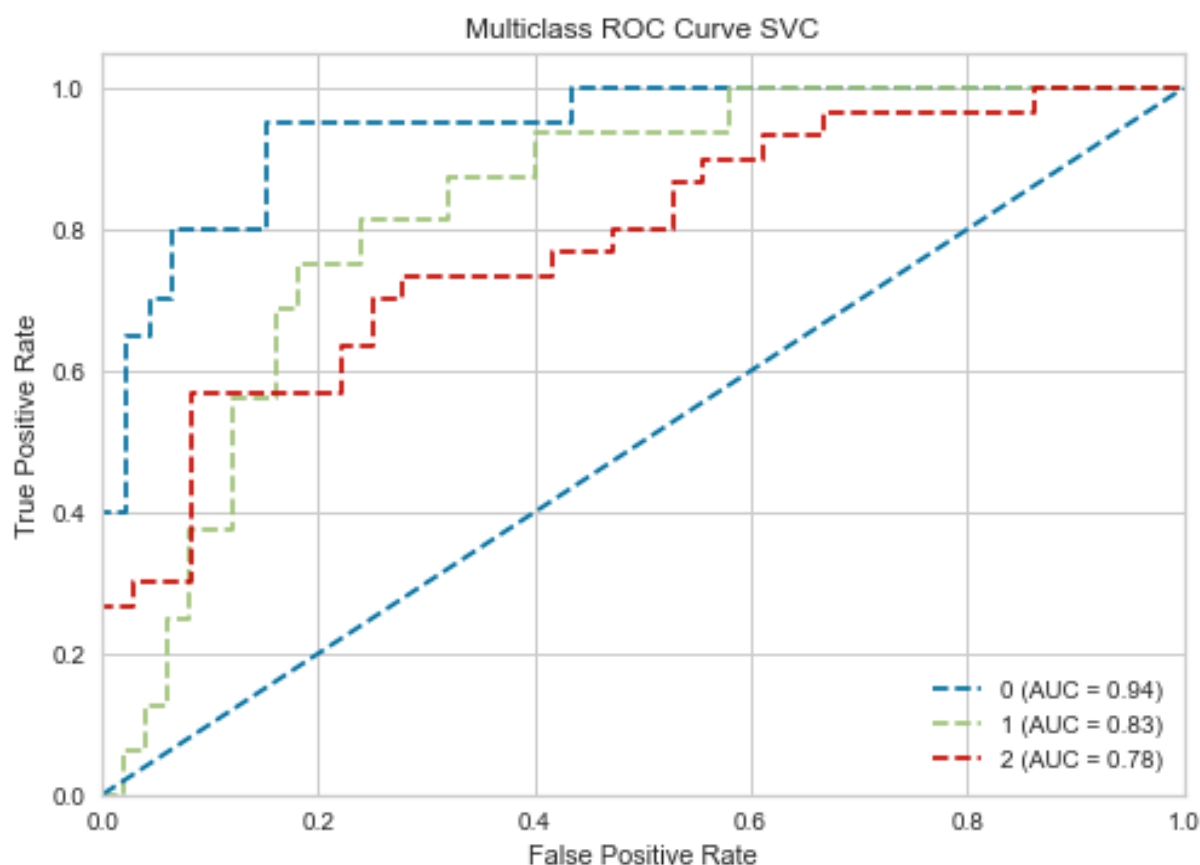


Fig 12.1: Classification report of Embedded method selected genes using SVC.

The precision and f1 values are higher for the CTL class, indicating better performance compared to the MCI and AD classes. The AD class presents lower recall, indicating more difficult predictions. However, the MCI class demonstrates moderate recall, suggesting relatively better prediction performance compared to AD.



Fig_12.2: ROC curve of Embedded method selected genes using SVC.

In this case, the CTL class achieves a high ROC-AUC score of 0.94, suggesting that the model performs well in distinguishing the CTL class from the other classes. The MCI class has a moderate ROC-AUC score of 0.83, indicating that the model's ability to discriminate the MCI class is relatively lower compared to CTL. The AD class has the lowest ROC-AUC score of 0.78, suggesting that the model's performance in differentiating the AD class from the other classes is relatively weaker.

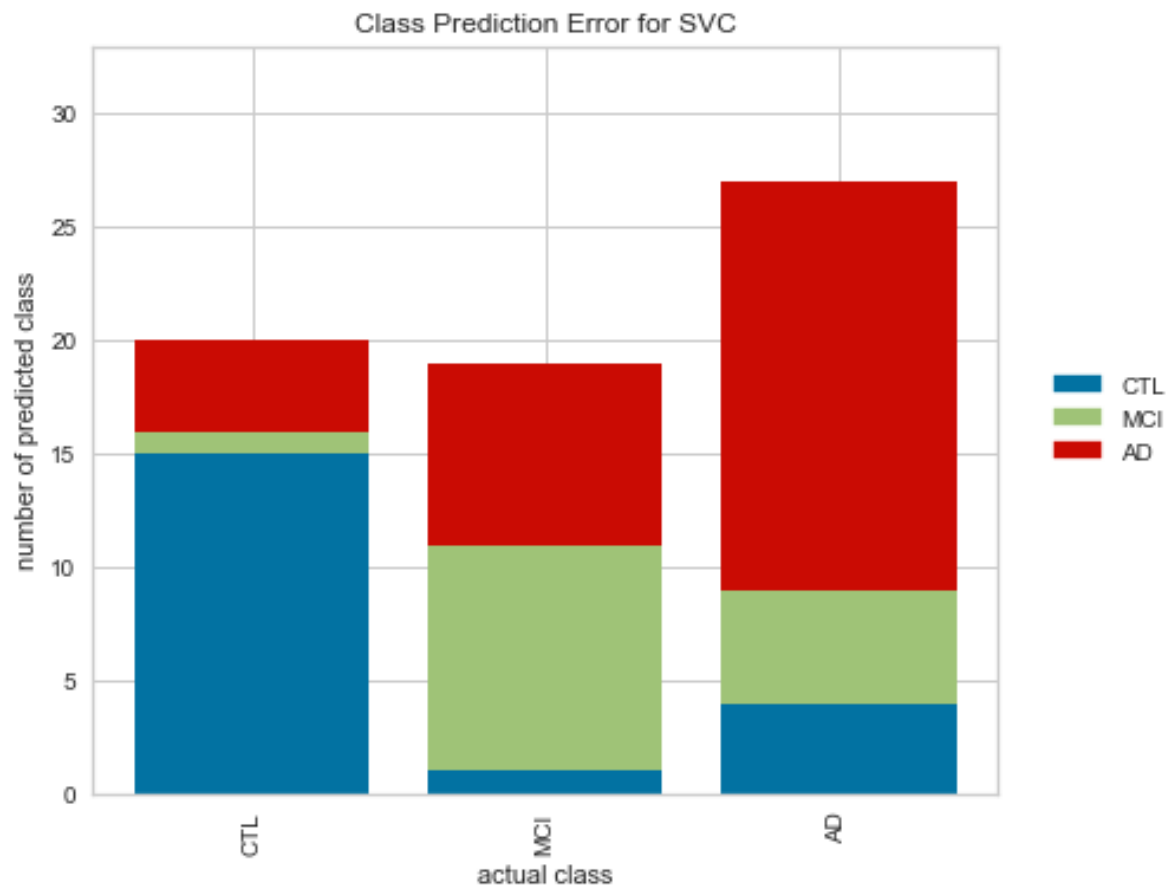
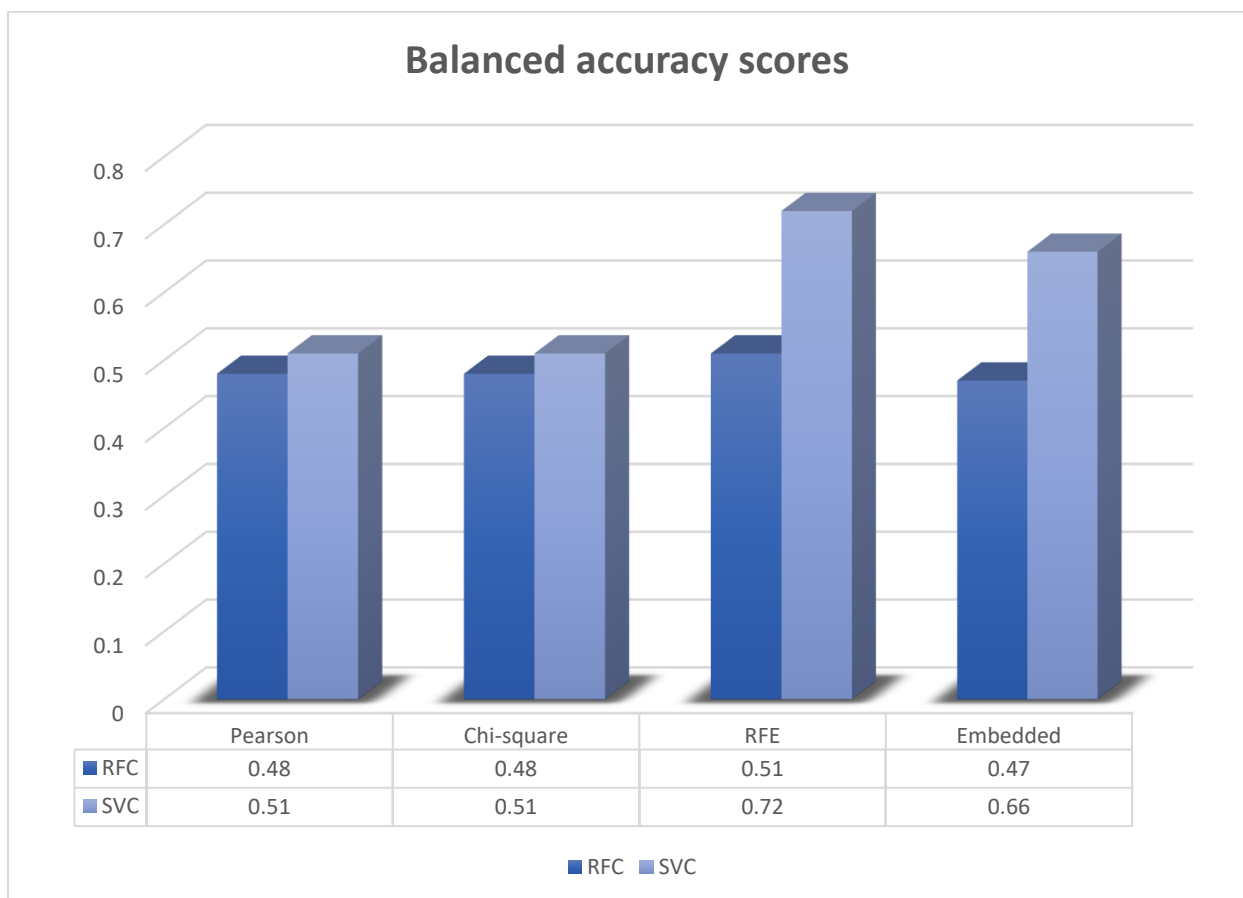


Fig 12.3: Class prediction error on Embedded method selected genes using SVC.

The SVC model, which utilized embedded method selected genes, exhibits accurate predictions for the CTL class. However, it encounters challenges when predicting the other classes, resulting in a higher confusion rate. This is evident from the comparison of class prediction error rates, where the CTL class has a lower error rate compared to MCI and AD classes.

Interpretation of Results:

After analysing the above graphs and conducting a thorough comparison of the results obtained through various feature selection techniques, it can be concluded that the features selected using Recursive Feature Elimination (RFE) with Support Vector Classifier (SVC) as the base estimator method exhibit superior performance. These selected features, when combined with the SVC classifier, demonstrate effective discrimination between the classes. The evaluation metrics, including BAS (balanced accuracy score) with a value of 72%, as well as Precision, Recall, F1-score, ROC-AUC score, and class prediction error rate, indicate that the SVC classifier, along with the chosen features, yields notable outcomes in predicting the disease stages.



Fig_13: Balanced accuracy scores of Trail_2.

Table_1: Comparison table of evaluation metrics of different feature selection methods.

		Pearson		Chi-square		RFE		Embedded	
		RFC	SVC	RFC	SVC	RFC	SVC	RFC	SVC
BAS		0.48	0.51	0.48	0.50	0.51	0.72	0.47	0.66
precision	CTL	0.64	0.77	0.67	0.79	0.62	0.89	0.62	0.75
	MCI	0.30	0.37	0.26	0.26	0.32	0.52	0.28	0.53
	AD	0.60	0.53	0.62	0.59	0.67	0.73	0.57	0.67
Recall	CTL	0.45	0.50	0.50	0.55	0.50	0.85	0.50	0.75
	MCI	0.50	0.44	0.44	0.38	0.50	0.69	0.50	0.62
	AD	0.50	0.60	0.50	0.57	0.53	0.63	0.40	0.60
F1-score	CTL	0.53	0.61	0.57	0.65	0.56	0.87	0.56	0.75
	MCI	0.37	0.40	0.33	0.31	0.38	0.59	0.36	0.57
	AD	0.55	0.56	0.56	0.58	0.59	0.68	0.47	0.63
ROC-AUC	CTL	0.81	0.82	0.82	0.88	0.81	0.96	0.82	0.94
	MCI	0.62	0.62	0.63	0.59	0.70	0.85	0.68	0.83
	AD	0.66	0.62	0.70	0.68	0.71	0.85	0.70	0.78

Table_2: Genes identified by SVC

Genes identified by RFE using SVC estimator	VPS9D1, UBA7, SNTB2, SHKBP1, SF3B6, RPS27A, RPL36AL, MRPL51, HSP90AA1, FES, DGKQ, COX7C, COMMD6, CMTM2, BOLA2, ATP6V1E1, ATP6V1D, ATP5J, ATP5EP2, ARPC3, APBB3, ABCA7, SORL1, SDF4, SBF1, RPS25, RASAL3, PSMA6, NDUFS5, NDUFA4, LY96, LDHB, DNAJA1, DCUN1D1, CREB5, CLDN15, C10orf54, ATP5L, ATP5I, ARHGAP4, RPS27, RAB37, PGGHG, NDUFB2, HELZ2, ATP5F1, ANKRD13D, AIF1, TBCA, SRP14, SLC15A4, CREBRF, ZNF444, USP21, UQCRBP1, TMEM50B, SRGN, SLC40A1, SF3B2, RNF13, PTEN, PRF1, POLR2C, PHC1, OR52N2, MFSD10, MEF2D, MAPKAP1, LRRC8B, JADE2, FAM212B, CSK, COX7B, COX16, CHKB, CCDC120, BTF3L4, APLP2, TOMM7, TNFAIP6, TAX1BP1, TAF1C, SNRPD2, SNHG5, S100A8, RWDD1, RPS3A, RPL39, RPL35, RPL17, RAB8B, PSD4, PFDN5, NBEAL2, LPAR2, LINC00936, ITGAL, ITGA2B, HINT1, FOXO3, EEF1B2, DPM1, CD3D, ANXA1, ADAM8, ACADVL
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Table_3: Few Mutual genes identified in Literature survey

The Predicted Key Molecules, Functions, and Pathways That Bridge Mild Cognitive Impairment (MCI) and Alzheimer's Disease (AD)(Tao et al., 2020)	SNRPD2, RPS27A, RPS3A, PFDN5, DNAJA1, ANXA1, S100A8
Bioinformatics analysis of diagnostic biomarkers for Alzheimer's disease in peripheral blood based on sex differences and support vector machine algorithm(Ji et al., 2022)	RPS27A, RPS3A, COX7C, PSMA6, PFDN5
JDSNMF: Joint Deep Semi-Non-Negative Matrix Factorization for Learning Integrative Representation of Molecular Signals in Alzheimer's Disease(Moon & Lee, 2021)	NDUFB2, ATP5J, ATP5F1, COX7C, NDUF5, COX7B
Robust Sampling of Defective Pathways in Alzheimer's Disease. Implications in Drug Repositioning(Fernández-Martínez et al., 2020)	MRPL51, RPL36AL, RPS25, RPL17, RPS27A, HSP90AA1, TAX1BP1, ABCA7
Meta-Analysis of Gene Expression Changes in the Blood of Patients with Mild Cognitive Impairment and Alzheimer's Disease Dementia(Bottero & Potashkin, 2019)	SNTB2, CMTM2, MRPL51, RPL36AL, NDUFB2, NDUF4A, AIF1, HELLZ2

4.DISCUSSION

This study aimed to use machine learning techniques to better understand the early stages of cognitive impairment. By analysing a gene expression dataset, the researchers applied machine learning algorithms to create predictive models and identify important genes related to cognitive impairment. The results obtained from this study provide valuable insights into how machine learning can help detect and diagnose cognitive impairment at an early stage.

The findings of this study show that machine learning can effectively predict cognitive impairment. The models developed exhibited reasonable accuracy, sensitivity, and specificity, suggesting that they can be valuable tools in clinical practice. Being able to accurately predict cognitive impairment can lead to early interventions and personalized treatment plans, ultimately improving the outcomes and quality of life for patients.

One of the major advantages of machine learning algorithms is their ability to analyse large datasets and identify complex patterns that may not be apparent using traditional statistical methods. In this study, the machine learning models identified several important genes associated with cognitive impairment. These findings support previous research and validate the models' ability to capture key factors contributing to cognitive impairment.

Additionally, this study highlights how machine learning can enhance the current diagnostic process for cognitive impairment. Traditional assessments heavily rely on subjective evaluations and clinical judgments, which can be influenced by various factors like differences among evaluators and cognitive biases. By incorporating machine learning models into the diagnostic process, clinicians can benefit from objective and standardized assessments, reducing the potential for errors and improving diagnostic accuracy.

Despite the promising results, it's important to acknowledge certain limitations. Firstly, the dataset used in this study was collected from a specific population, which may limit the generalizability of the findings to other demographics or settings. Future research should include more diverse populations to ensure that machine learning models are robust and applicable across different contexts.

Secondly, while machine learning models provide accurate predictions, understanding how these models reach their decisions remains a challenge. It is crucial to comprehend the underlying mechanisms and decision-making processes of machine learning algorithms for their clinical implementation. Future studies should focus on developing models that are interpretable and provide insights into the features and patterns contributing to predicting cognitive impairment.

Another limitation is the reliance on retrospective data, which introduces potential biases and hampers the ability to establish causal relationships. Prospective studies that collect data over time and investigate the predictive value of machine learning models in longitudinal settings would provide stronger evidence for the effectiveness of these techniques.

In brief, this study demonstrates the potential of machine learning in studying cognitive impairment. Accurate prediction and identification of key genes associated with cognitive impairment can facilitate early detection and personalized interventions. However, further research is needed to address the mentioned limitations and ensure that machine learning models are robust, generalizable, and interpretable in clinical practice. With advancements in machine learning and access to larger and more diverse datasets, this field shows promise for revolutionizing our understanding and management of cognitive impairment.

5.CONCLUSION

Alzheimer's disease presents a significant healthcare concern, and rather than seeking a definitive cure, the emphasis lies in minimizing risk, implementing early interventions, and achieving accurate and timely diagnosis of symptoms. Extensive literature surveys have revealed numerous endeavours in utilizing various machine learning algorithms and micro-simulation methods for the detection of Alzheimer's Disease.

Nevertheless, the identification of pertinent attributes capable of early detection remains a formidable undertaking. Subsequent research will prioritize the extraction and analysis of novel genes that exhibit greater potential in aiding the detection of Alzheimer's Disease. Additionally, efforts will be directed towards eliminating redundant and irrelevant features from existing gene sets, aiming to enhance the accuracy of detection techniques. Incorporating metrics such as log loss and kappa score, as well as incorporating Education as a variable in our model, will enable effective training to differentiate between healthy individuals and those afflicted with Alzheimer's.

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